IRRATIONALITY MEASURES FOR DIFFERENCES OF TRIGAMMA VALUES

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ABSTRACT. For $0 < s < t \le 1$ let $\Psi(s,t) = \psi'(s) - \psi'(t)$, a difference of values of the trigamma function. Catalan's constant is $G = \frac{1}{16}\Psi(\frac{1}{4}, \frac{3}{4})$. Building on OpenAI's determinant method for the irrationality of G, we prove irrationality measures for $\Psi(\frac{k}{q}, 1) = \psi'(\frac{k}{q}) - \zeta(2)$ with $q \in \{3,4,5,7\}$ and $\gcd(k,q) = 1$, and for $\Psi(\frac{1}{3}, \frac{2}{3}) = 9L(2, \chi_{-3})$. In classical terms, with μ the irrationality exponent,

$$\mu\Big(G \pm \frac{5\pi^2}{48}\Big) \le 31.2876, \qquad \mu\big(L(2,\chi_{-3})\big) \le 13.7046,$$
$$\mu\Big(L(2,\chi_{-3}) \pm \frac{\pi^2}{9}\Big) \le 9.4998,$$
$$\mu\big(\psi'(\tfrac{k}{5}) - \zeta(2)\big) \le 23.8272 \ (1 \le k \le 4),$$
$$\mu\big(\psi'(\tfrac{k}{7}) - \zeta(2)\big) \le 2207.8697 \ (1 \le k \le 6).$$

For $L(2,\chi_{-3})$ and $L(2,\chi_{-3}) \pm \pi^2/9$, which Calegari, Dimitrov and Tang proved irrational, the previous best bound we know of is their effective bound $\mu \le 24781$. We know of no earlier irrationality measure for the other numbers, and no earlier irrationality result for $\psi'(\frac{k}{5}) - \zeta(2)$ and $\psi'(\frac{k}{7}) - \zeta(2)$. The same construction at $s = \frac{1}{2}$ gives $\mu(\pi^2) \le 6.5253$; this is weaker than the known bounds and is included to show the reach of the method.

The approximations come from determinants of *rectangular* Löwner matrices of digamma differences at integer nodes, with a long common block and an interleaved border whose length is the only parameter of the geometry. The denominators are bounded prime by prime by a finite list of local bounds (assignment problems, the rank of the reduction modulo p, and windows of the border free of poles, where the reduction is translation invariant; for the small primes, which meet the border several times, the windows are read modulo p), each an explicit integer expression in the scale, the border, the prime and the first poles of the frame modulo p. For the primes at which the border meets a residue class more than once, a bound at third order in the Smith normal form comes from eliminating the pairs of border nodes in one class by Schur complements and separating the remaining border nodes by rational interpolation; one theorem turns a finite table of affine pieces, checked by comparisons of rational numbers, into the content estimate for every shift with the same denominator q. Nonvanishing follows from Hermite interpolation in characteristic zero, after deleting a bounded number of common nodes. The size is bounded with Andréief's identity and a lower bound for a discrete logarithmic energy, obtained from a comparison measure and verified on a finite cover by cells.

The proof is formalized in Lean 4, with no unproved steps and only the standard axioms.

CONTENTS

1. Introduction 3
2. Overview of the method and of the new ideas 6
3. The construction 10

4. Arithmetic: the content 12
5. Nonvanishing 25
6. Analytic decay 26
7. The number $L(2,\chi_{-3})$ 30
8. Proof of Theorem 1.1 32
9. The families 33
10. Limits of the method 34
11. Formalization 35
Disclosure of AI use 36
References 36

1. Introduction

In October 2026 OpenAI [31] proved that Catalan's constant

$$G=\sum_{j=0}^{\infty}\frac{(-1)^j}{(2j+1)^2}=0.91596\,55941\ldots$$

is irrational, and formalized the proof in Lean. (An earlier claimed proof [38] was found to contain a gap [41].) Their argument, and the irrationality-measure results built on it [17, 9, 24], use a determinant of large size whose denominators and decay are balanced very closely.

In the present paper we study a different, much smaller, family of determinants: rectangular Löwner matrices of digamma differences. They approximate the differences of values of the trigamma function ψ',

$$\Psi(s,t)=\psi'(s)-\psi'(t)=\sum_{n=0}^{\infty}\Big(\frac{1}{(n+s)^2}-\frac{1}{(n+t)^2}\Big),\qquad 0<s<t\le 1.$$

Catalan's constant is one of these numbers, $G=\frac{1}{16}\Psi(\frac14,\frac34)$, and the companion paper [24] treats it with OpenAI's construction. Here the second shift is $t=1$, with one exception, $L(2,\chi_{-3})=\frac19\Psi(\frac13,\frac23)$, where

$$L(2,\chi_{-3})=\sum_{n\ge0}\Big(\frac{1}{(3n+1)^2}-\frac{1}{(3n+2)^2}\Big)=0.78130\,24128\ldots$$

For $t=1$ and $s=\frac{k}{q}$ the value $\Psi(\frac{k}{q},1)=\psi'(\frac{k}{q})-\zeta(2)$ is an L-value or a combination of L-values plus a rational multiple of π^2; for $q=4$ and $q=3$,

$$\Psi(\tfrac14,1)=8\Big(G+\frac{5\pi^2}{48}\Big),\qquad \Psi(\tfrac34,1)=-8\Big(G-\frac{5\pi^2}{48}\Big),$$
$$\Psi(\tfrac13,1)=\frac92\Big(L(2,\chi_{-3})+\frac{\pi^2}{9}\Big),\qquad \Psi(\tfrac23,1)=-\frac92\Big(L(2,\chi_{-3})-\frac{\pi^2}{9}\Big).$$

We also write $\lambda_\pm=\frac{G}{2}\pm\frac{5\pi^2}{96}$, so that $G\pm5\pi^2/48=2\lambda_\pm$, $\Psi(\frac14,1)=16\lambda_+$ and $\Psi(\frac34,1)=-16\lambda_-$.

Recall that the *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large b. It does not change under $\xi \mapsto r\xi + r'$ with $r, r' \in \mathbb{Q}$, $r \neq 0$, so a bound for a value of Ψ is a bound for the classical constant in the same row of Table 1.

(s,t)	the number, up to a rational affine map	B
$(\frac{1}{4},1)$, $(\frac{3}{4},1)$	$G + \frac{5\pi^2}{48}$, $G - \frac{5\pi^2}{48}$	31.2876
$(\frac{1}{3},\frac{2}{3})$	$L(2,\chi_{-3})$	13.7046
$(\frac{1}{3},1)$, $(\frac{2}{3},1)$	$L(2,\chi_{-3}) + \frac{\pi^2}{9}$, $L(2,\chi_{-3}) - \frac{\pi^2}{9}$	9.4998
$(\frac{k}{5},1)$, $k = 1,2,3,4$	$\psi'(\frac{k}{5}) - \zeta(2)$	23.8272
$(\frac{k}{7},1)$, $k = 1,\dots,6$	$\psi'(\frac{k}{7}) - \zeta(2)$	2207.8697
$(\frac{1}{2},1)$	π^2	6.5253

TABLE 1. The pairs of shifts and the exponents of Theorem 1.1.

Theorem 1.1. *Let (s,t) and B be in one row of Table 1. For every $\varepsilon > 0$ there is b_0 such that for all coprime integers a, b with $b \geq b_0$,*

$$\left|\Psi(s,t) - \frac{a}{b}\right| > b^{-B-\varepsilon}.$$

In particular $\Psi(s,t)$ and the number in the same row are irrational, with irrationality exponent at most B.

The rows are one construction with different shifts. The frame is the digamma difference $\psi(z+t)-\psi(z+s)$, or for $L(2,\chi_{-3})$ a combination of three of them; for the pairs $(\frac{k}{q},1)$ the node geometry depends on q only through one rational parameter. The last row is the shift $s = \frac{1}{2}$, with $\Psi(\frac{1}{2},1) = \psi'(\frac{1}{2}) - \zeta(2) = 2\zeta(2) = \pi^2/3$. It is not a new result: the bound 6.5253 is weaker than the known bounds for $\mu(\pi^2) = \mu(\zeta(2))$, namely 5.0193784 of Giraudin [15] and 4.62675 of Leibtag et al. [23]. We include it because it shows the reach of the construction: one family of determinants, with the same three estimates, covers π^2, the numbers $G \pm 5\pi^2/48$, $L(2,\chi_{-3})$ and $L(2,\chi_{-3}) \pm \pi^2/9$, and the differences $\psi'(\frac{k}{q}) - \zeta(2)$ for $q = 5, 7$. Theorem 1.1, together with every theorem, proposition and lemma used in its proof, has been verified in the Lean 4 proof assistant, with no unproved steps and only the standard axioms; see Section 11. The remarks and Section 10, which discuss the method, are not part of the formal proof. The formalization is available at `https://github.com/slitsey/catalan-irrationality-measure`. The use of AI in this work is disclosed at the end of the paper.

1.1. **Previous results.** For earlier work on the arithmetic of G and related values see [36, 45, 44, 20, 21] and the survey [48]; for recent rational approximations to G see Eskandari [11]. Finite irrationality exponents for G itself were obtained after OpenAI's proof by Jha [17], Carreira [9], Bojnal [4],

Xu [42] and the author [24]; Carreira [8] also records attempts to carry the determinant method to $\beta(4)$. Our searches found no previous irrationality result, and no irrationality measure, for $G - 5\pi^2/48$ or $G + 5\pi^2/48$. Eskandari, Murty and Nemoto [12] construct a 2-dimensional mixed motive over $\mathbb{Q}$ with period G and use it to obtain linear forms in 1 and G. Eskandari [11] improves the approximation exponent of these Rivoal–Zudilin type forms from 0.52 to 0.62 (a preprint; explicit approximations, not a lower bound for all rationals). Sun's preprint [38] claimed a proof of the irrationality of G; Wachs [41] reports that its main estimate is not justified.

Calegari, Dimitrov and Tang proved that 1, $\zeta(2)$ and $L(2,\chi_{-3})$ are linearly independent over $\mathbb{Q}$ [6], by a new arithmetic holonomy bound applied to a construction of Zagier. In [7] they made holonomy bounds effective; Theorem 5.1 there gives $\mu(L(2,\chi_{-3})) \le 24781$. The remark following that theorem states that the same theorem and proof hold for any $\mathbb{Q}$-linear combination of $L(2,\chi_{-3})$ and $\zeta(2)$; in particular 24781 was the known bound for $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$. Theorem 1.1 improves it to 13.7046 for $L(2,\chi_{-3})$, and to 9.4998 for the two numbers $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$. Leibtag et al. [23] find numerically that Apéry-like approximations from conservative matrix fields have positive approximation exponents for $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$, and a supplement to their paper uses one such field to give an Apéry-style proof of the theorem of Calegari, Dimitrov and Tang; the exponent it proves for each of these two numbers is only greater than 10^{-4}. For the numbers $\psi'(\frac{k}{5}) - \zeta(2)$ and $\psi'(\frac{k}{7}) - \zeta(2)$ of Theorem 1.1 our searches found no earlier irrationality result; the irrationality of $\mathrm{Li}_2(1/5)$ proved by Prellberg [33] concerns a different number.

For comparison, among classical constants, the bound $\mu(\log 2) < 3.5746$ of Marcovecchio [28], the bounds of Rhin and Viola for $\zeta(2)$ and $\zeta(3)$ [34, 35] (after Apéry and Beukers [1, 3]) and Zudilin's bound for $\zeta(2)$ [46] have been joined, since September 2026, by a series of new results, several of them obtained with AI systems. OpenAI proved $\mu(\pi) = 2$, with a Lean formalization [32], improving the bound $\mu(\pi) < 7.1033$ of Zeilberger and Zudilin [43]. Liu, Jiang and Zhang proved $\mu(\log r) = 2$ for every rational $r > 0$, $r \ne 1$, and for nonzero rational combinations of such logarithms [25]. Giraudin lowered the bound for $\mu(\zeta(2))$ to 5.0193784 with two hypergeometric constructions of Zudilin [46] at new parameters [15], and the bound for $\mu(\zeta(3))$ to 5.5138800 by applying the Rhin–Viola method at a new integral direction [16]. Leibtag, Barkan, Kalisch, Weinbaum, Shalyt and Kaminer [23] showed that the classical constructions are one-dimensional trajectories in conservative matrix fields, and by searching over other trajectories obtained $\mu(\zeta(2)) < 4.62675$ and $\mu(\zeta(3)) < 5.4197$. Prellberg [33] proved that $\mathrm{Li}_2(-1/4)$, $\mathrm{Li}_2(1/5)$ and $\mathrm{Li}_2(-1/3)$ are irrational, three cases left open in [6], with the integrals of Rhin–Viola and Viola–Zudilin. Fauzan [13] announced a proof that $\zeta(5)$ is irrational, using rationally normalized Hankel determinants, and Lai [22] combined his method with simultaneous approximations of Zudilin and of

Dauguet and Zudilin to show that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$.

Related work on determinants. Irrationality proofs built on determinants of large size have a precedent in Zudilin's determinantal criterion with Hankel determinants [47]; the preprint of Liu, Jiang and Zhang on logarithms [25] uses a separated-weight interpolation determinant, and the arguments of Fauzan and Lai [13, 22] use Hankel determinants. Brown [5] proposes a multivariate transfinite-diameter test for Mellin-type integrals, applied to $\zeta(2)$. The framework we use, determinants whose decay is controlled by Cauchy determinants and Andréief's identity and whose denominators are controlled prime by prime, is OpenAI's [31]. Our determinants are Löwner (divided-difference) matrices of a digamma difference, a matrix-valued analogue of Padé and Hermite–Padé approximation [30]. Formal proofs of irrationality results in Lean also include the irrationality of $\zeta(3)$ by Liu, Zhang and Zhi [26].

1.2. **Outline of the proof.** Section 2 explains the construction and the new ideas without technicalities, and states the general deduction. Section 3 gives the construction; Sections 4, 5 and 6 prove the three estimates (content, nonvanishing, decay), uniformly in the shift, and Section 7 treats the frame of $L(2,\chi_{-3})$. Section 8 deduces Theorem 1.1 and lists the data of each row, Section 9 collects the facts specific to each family, and Section 10 discusses the limits of the method. Section 11 describes the formalization, and the disclosure of AI use follows it.

1.3. **Notation and conventions.** ψ is the digamma function, the logarithmic derivative of Euler's gamma function; v_p is the p-adic valuation; for $P \in \mathbb{Q}[x]$, $\operatorname{den} P$ is the least positive integer whose product with P lies in $\mathbb{Z}[x]$; $[0,r) = \{0,1,\dots,r-1\}$. For finite sets I, J of integers and a matrix A indexed by integers, $A[I,J]$ is the submatrix with rows I and columns J.

2. Overview of the method and of the new ideas

2.1. **Löwner matrices of digamma differences.** For a function F analytic near a finite set of distinct points, the *Löwner matrix* [27] is the matrix of divided differences, indexed by the points:

$$\mathsf{L}_F[t,t'] = \frac{F(t') - F(t)}{t' - t} \ (t \neq t'), \qquad \mathsf{L}_F[t,t] = F'(t).$$

It is a Cauchy-like structured matrix, and Hermite's formula gives $\mathsf{L}_F = \frac{1}{2\pi i} \oint F(z)\, v(z) v(z)^{\top} dz$ with $v_t(z) = 1/(z-t)$, so that a representation of F as a Stieltjes or contour integral exhibits L_F as a Gram matrix of Cauchy kernels.

For a rational shift $0 < s < 1$ with denominator q we take the digamma difference F_s, and for $L(2, \chi_{-3})$ a combination F_χ of three digamma functions, with integer nodes:

$$\begin{gathered} F_s(z) = \psi(z+1) - \psi(z+s), \qquad \omega = \tfrac{1}{q^2}, \quad \xi_s = -\tfrac{1}{q^2}\Psi(s,1), \\ F_\chi(z) = -\tfrac{1}{3}\psi(z+\tfrac{1}{3}) + \tfrac{1}{2}\psi(z+\tfrac{1}{2}) - \tfrac{1}{6}\psi(z+1), \\ \omega = -\tfrac{1}{6}, \quad \xi_\chi = \tfrac{1}{4}L(2,\chi_{-3}). \end{gathered} \tag{2.1}$$

In each case $\xi = \omega\, F'(0)$, while $F(t') - F(t)$ and $F'(t) - F'(0)$ are rational for integers $t, t' \geq 0$. Hence, for any finite sets of nonnegative integer nodes R (rows) and Q (columns),

$$\omega \,\mathsf{L}_F[R,Q] = M[R,Q] + \xi\, E_{R,Q}, \qquad E_{R,Q}[t,t'] = \mathbf{1}[t = t'], \tag{2.2}$$

with an explicit rational symmetric kernel M on $\mathbb{Z}^2_{\geq 0}$ (§3). Replacing ξ by a variable x gives a matrix polynomial $\Phi(x) = M[R,Q] + xE_{R,Q}$, whose determinant $P(x)$ is a polynomial with rational coefficients of degree at most $|R \cap Q|$, and $P(\xi) = \det(\omega\,\mathsf{L}_F[R,Q])$. If $|\xi - a/b|$ is very small, $P(a/b)$ is very close to $P(\xi)$, which is tiny; on the other hand $b^{|R\cap Q|}\operatorname{den}(P)\,P(a/b)$ is an integer. If it is not zero we get a contradiction as soon as

$$\underbrace{\log \operatorname{den} P}_{\text{content}} + |R \cap Q| \log b \;<\; \underbrace{-\log|P(a/b)|}_{\text{decay}}.$$

Three estimates are therefore needed: an upper bound for the *content* $\log \operatorname{den} P$, an upper bound for $|P(x)|$ near ξ (the *decay*), and *nonvanishing* of $P(a/b)$ for some member of a family of determinants of a size that can be chosen in terms of b. Each estimate is proved once, for every shift, and the families differ only in a few parameters.

2.2. **New idea 1: rectangular frames with an interleaved border.** We take a long common block $C = [0, r)$, which carries the variable x, and a border of $m \approx \alpha r$ row nodes and m column nodes, interleaved:

$$X' = \{r, r+2, \ldots, r+2m-2\} \text{ (rows only)},$$
$$Y' = \{r+1, r+3, \ldots, r+2m-1\} \text{ (columns only)},$$

so that $R = C \sqcup X'$ and $Q = C \sqcup Y'$. The determinant $P(x) = \det \Phi(x)$ still has degree at most r, so the integrality loss $|R \cap Q| \log q$ is that of r nodes, but the matrix has size $r + m$ and its decay is that of a configuration of $(1+\alpha)r$ nodes. The content grows too, but less: the border rows and columns never meet the diagonal. The *members* of the family are the principal restrictions obtained by deleting a bounded number of common nodes. The single parameter of the geometry is $c = 2\alpha$, the length of the border relative to r; we use $c = \frac{3}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{9}, \frac{1}{6}$ for $q = 2, 4, 3, 5, 7$ and $c = \frac{1}{2}$ for $L(2, \chi_{-3})$. The interleaving of X' and Y' is essential: it keeps the two Cauchy determinants in Andréief's formula comparable (§6), and it gives the reduction of the matrix modulo a prime the alternating structure on which several of the arithmetic bounds of §4 rest.

2.3. **New idea 2: the content from a finite list of local bounds.** As in the work of Rhin and Viola [34, 35], we bound the p-adic valuation of den P prime by prime and sum the bounds against $\log p$ with the prime number theorem in arithmetic progressions. For a prime p in the range $p \asymp r$ the relevant data are the scale r, the border size m, the prime, and the first index $k_p \geq 1$ at which the second digamma term of F_s has a pole modulo p; we have $k_p = \kappa p + O(1)$ with $\kappa = j/q$, where $j \equiv qs\,p^{-1} \pmod q$ is the *root class* of p. The frame of $L(2, \chi_{-3})$ has two such first poles, and on each class of primes modulo 6 both are affine functions of one index of the same kind (Section 7).

We prove a finite list of local bounds, each by its own mechanism: the assignment (tropical) bound with explicit potentials, three bounds for the rank of the reduction modulo p of a rescaled matrix, and a bound from windows of consecutive border nodes free of poles, in which the reduction is translation invariant between residue classes. The window bound also applies to the primes $p \leq m$, which meet the border several times: reduced modulo p, the border covers an arc of residue classes once more than the others, and on this arc the translation argument improves the assignment bound. Every one of them is an explicit integer expression in (r, m, p, k_p), built from affine forms with min, max, truncated subtraction, the distance to $p\mathbb{Z}$ and the multiple $p\lfloor r/p \rfloor$, and holds in a range of p/r described by linear inequalities. Replacing (r, m, p, k_p) by $(1, \alpha, \theta, \kappa\theta)$ turns each bound into a piecewise affine profile in $\theta = p/r$, up to an error bounded by an explicit constant. The content constant is then

$$\Lambda = \frac{1}{\varphi(q)} \sum_j \int_0^\infty \mathcal{P}_j(\theta)\, d\theta,$$

where $\mathcal{P}_j$ is the pointwise minimum of the valid profiles for the root class j. A finite table of affine pieces above $\mathcal{P}_j$, together with the name of the bound that justifies each piece, is a certificate for Λ: whether it is correct is decided by finitely many comparisons of rational numbers. One theorem (Theorem 4.14) turns an accepted table into the content estimate. The bounds depend on the shift s only through the root class, so the same table serves every shift with denominator q, and the numbers $\psi'(s) - \zeta(2)$ for these shifts receive the same exponent.

2.4. **New idea 3: the content at third order when the border wraps around.** For the primes $p \leq 2m$ the border $[r, r + 2m)$ meets some residue classes modulo p more than once, and the bounds above, which work at first or second order in the Smith normal form of the rescaled matrix, are not sharp there. We gain at third order. Write $m = np + e$ with $0 \leq e < p$ and $d = \|m\|_p$. After the common block is eliminated, each residue class that carries a pair of border nodes ℓ, $\ell + p$ (one row, one column) contributes a unit pivot: its leading coefficient is the increment of the frame over $(\ell, \ell + p]$, which contains exactly one pole of each digamma term, and the two cannot cancel.

Eliminating these pivots by Schur complements leaves a matrix whose Smith normal form looks like that of a geometry with a shorter common block and a border of d rows and d columns, and the separation argument of the bound (S) can be run on it at third order. Two facts make this work. First, the Schur complement changes the second-order term, but the change is exactly compensated by replacing the moments of the common block by the moments of the mixed (nonsymmetric) blocks that the elimination produces; this is a pair of identities for bordered inverses. Second, the remaining border nodes must still be separated from each other. For a cut at the second pole we deform the shift, $s \mapsto s+pY$, and evaluate at the pole, where the bordered system becomes a pair of Cauchy blocks. For a cut at a multiple of p we take the leading coefficient in Y, which leaves the harmonic numbers; there the bordered system is a problem of rational interpolation of type (n, n) on consecutive integers, its value at infinity is $H_{\beta+n} + 2H_n$ (with β the number of common nodes of the class), and two neighbouring classes differ by $1/(\beta+n+1)$, a p-adic unit. The resulting bound (U) is again an explicit integer expression in (r, m, p, k_p), and it holds for every member with no loss for the deleted nodes.

2.5. **New idea 4: nonvanishing by Hermite interpolation in characteristic zero.** The kernel of $\Phi(x_0)$ at a rational point x_0 is identified with a space of pairs of polynomials satisfying Hermite interpolation conditions at the nodes. Two such pairs have vanishing cross product by a count of degrees and zeros, and a further pair in the kernel would give a rational antidifference of the increment of the frame, which does not exist because the pole at -1 is isolated in its class modulo $\mathbb{Z}$. This bounds the corank of $\Phi(x_0)$ by a number depending only on the frame and the period of the geometry, and a principal restriction deleting at most that many common nodes is nonsingular (Theorem 5.1). No prime enters, so the size r can be chosen freely in terms of b.

2.6. **New idea 5: decay through a certificate measure checked on a finite cover.** Moving Hermite's contour to a polygonal path Γ in the left half-plane writes $\omega\, \mathsf{L}_F[R, Q]$ as a Gram matrix $\int v_t v_{t'}\, dm$ of a complex measure m (signed atoms at poles of F plus an exponentially small density on Γ). Andréief's identity with two families of functions (rows and columns) gives

$$|\det \mathsf{L}_F[R, Q]| \le \frac{1}{n!} \int |\mathsf{C}_R(z)|\, |\mathsf{C}_Q(z)| \prod_{i=1}^{n} d|m|(z_i),$$

where $\mathsf{C}_R, \mathsf{C}_Q$ are Cauchy determinants. Taking logarithms turns the integrand into a discrete logarithmic energy of the free points z_i against the nodes in an external field. We bound this energy from below by comparing it with a *certificate measure* σ, a density that is polynomial on each edge of finitely many segments, through the conditional positive definiteness of the logarithmic kernel. What is left is a lower bound for one explicit function

g on the contour, on the segment carrying the atoms and on the interval of the nodes. We prove it on a finite cover of these sets by cells, each checked in exact rational arithmetic. One theorem (Theorem 6.2) gives the decay for every geometry parameter $c \le 1$ and every frame whose poles are separated from the contour; a family supplies only σ, the polygon and the cover.

2.7. **The deduction.** Write $\eta = D - \Lambda$ and let γ be the window constant of the decay estimate: the decay holds at every point within $e^{-\gamma r}$ of ξ. The three estimates are combined in §8; the result is

$$\mu(\xi) \le \frac{\gamma}{\eta} = \frac{\gamma}{D - \Lambda}.$$

Table 2 lists the constants for every row of Table 1.

2.8. **Formal verification.** Every step, including the local arithmetic bounds and the checker of the content tables, the interpolation argument for nonvanishing, the contour representation, the cell checks of the decay certificates and the prime number theorem in arithmetic progressions, is formalized in Lean 4 with Mathlib [10, 39]. The formal proofs are built on a shared core (rectangular determinants, Gauss valuations, the local bounds, the content checker, the characteristic-zero selection, the decay theorem and the measure deduction); a family supplies its frame, its geometry parameter and its data tables. The written proofs below are at the level of a research article; for long computations we point to the formal proof, which is the basis for the correctness of Theorem 1.1.

3. The construction

3.1. **The kernels.** Let $0 < s < 1$ be rational with denominator q, and put $o = q(1 - s)$, so that $k - 1 + s = (qk - o)/q$. For the function F_s of (2.1) the kernel M of (2.2) is as follows; here $0 \le t < t'$ are integers and $M[t', t] = M[t, t']$:

$$M[t, t'] = \frac{1}{q^2(t' - t)} \sum_{k=t+1}^{t'} \Big(\frac{1}{k} - \frac{q}{qk - o} \Big),$$

$$M[t, t] = -\frac{1}{q^2} \sum_{k=1}^{t} \Big(\frac{1}{k^2} - \frac{q^2}{(qk - o)^2} \Big).$$

For $s = \frac{3}{4}$ this is $M[t, t'] = -\frac{1}{t' - t} \sum_{k=t+1}^{t'} \frac{1}{16k(4k-1)}$. For the function F_χ,

$$M[t, t'] = \frac{1}{6(t' - t)} \sum_{k=t+1}^{t'} g(k), \qquad M[t, t] = \frac{1}{6} \sum_{k=1}^{t} g'(k),$$

with $g(k) = \frac{1}{3k-2} - \frac{1}{2k-1} + \frac{1}{6k}$, so that $-\frac{1}{6} g'(k) = \frac{1}{2(3k-2)^2} - \frac{1}{3(2k-1)^2} + \frac{1}{36k^2}$.

Lemma 3.1. *For each function of (2.1), $\omega \mathbin{\mathsf{L}}_F [R, Q] = M[R, Q] + \xi E_{R,Q}$ for all finite sets $R, Q \subset \mathbb{Z}_{\ge 0}$, with ω and ξ as in (2.1).*

Proof. From $\psi(w+1)-\psi(w)=1/w$ one gets $F_s(k)-F_s(k-1)=\frac{1}{k}-\frac{1}{k-1+s}=\frac{1}{k}-\frac{q}{qk-o}$, and telescoping gives the off-diagonal entries. Similarly $\psi'(w+1)-\psi'(w)=-1/w^2$ gives $F'_s(t)-F'_s(0)$. Finally $\omega F'_s(0)=q^{-2}(\psi'(1)-\psi'(s))=-q^{-2}\Psi(s,1)=\xi_s$. For F_χ use $\psi'(\frac{1}{2})=\frac{\pi^2}{2}$ and $\psi'(\frac{1}{3})=\frac{2\pi^2}{3}+\frac{9}{2}L(2,\chi_{-3})$, which give $-\frac{1}{6}F'_\chi(0)=-\frac{1}{6}\sum_{k\ge1}g'(k)=\frac{1}{4}L(2,\chi_{-3})$. □

Since $\mu(u\xi+u')=\mu(\xi)$ for rational $u\neq0$ and u', Theorem 1.1 is a statement about the numbers ξ of (2.1).

3.2. **Geometries.**

Definition 3.2. Let $r,m\ge0$ and $e\in\{0,1\}$. The *geometry* $\mathcal{G}(r,m,e)$ consists of the common nodes $C=[0,r)\cup\{r+2m:\ e=1\}$, the row nodes $X'=\{r+2i:0\le i<m\}$ and the column nodes $Y'=\{r+2i+1:0\le i<m\}$. Rows are indexed by $R=C\sqcup X'$, columns by $Q=C\sqcup Y'$. For a set $I\subseteq C$ of deleted common nodes, the matrix

$$\Phi_{\mathcal{G},I}(x)=\big(M[t,t']+x\,\mathbf{1}[t=t'\in C\setminus I]\big)_{t\in R\setminus I,\ t'\in Q\setminus I}$$

is square of size $|C\setminus I|+m$; put $P_{\mathcal{G},I}(x)=\det\Phi_{\mathcal{G},I}(x)\in\mathbb{Q}[x]$, and write $\Phi_{\mathcal{G}}$, $P_{\mathcal{G}}$ when $I=\emptyset$.

By Lemma 3.1, $\Phi_{\mathcal{G}}(\xi+h)=\omega\,\mathsf{L}_F[R,Q]+h\,E_{R,Q}$ for real h. The geometry with $m=0$, $e=0$ is the square Löwner matrix of size r.

Definition 3.3. Let $0<c\le1$ and $\alpha=c/2$. The *alternating geometry* of parameter c at an even scale r is $\mathcal{G}(r,m(r),0)$ with $m(r)=\lceil\alpha r-\frac{1}{2}\rceil$, so that $|m(r)-\alpha r|\le\frac{1}{2}$ and the border $[r,r+2m)$ has length $cr+O(1)$. For a depth $K\ge0$, its *members* are the matrices $\Phi_{r,I}=\Phi_{\mathcal{G},I}$ with $|I|\le K$, and $P_{r,I}=\det\Phi_{r,I}$.

For the frames F_s we use $c=\frac{3}{10},\frac{1}{5},\frac{3}{10},\frac{2}{9},\frac{1}{6}$ for $q=2,4,3,5,7$, and for F_χ we use $c=\frac{1}{2}$ (Section 7). All $o(1)$ terms below are uniform over the members as $r\to\infty$.

Lemma 3.4 (integrality)**.** *$P_{\mathcal{G},I}$ has degree at most $|C\setminus I|$. If a/b is a fraction in lowest terms with $P_{\mathcal{G},I}(a/b)\neq0$, then $|\operatorname{den}(P_{\mathcal{G},I})\,P_{\mathcal{G},I}(a/b)|\ge b^{-|C\setminus I|}$.*

Proof. The variable occurs only in the $|C\setminus I|$ diagonal entries indexed by $C\setminus I$, so each Leibniz term has degree at most $|C\setminus I|$. Then $b^{|C\setminus I|}\operatorname{den}(P_{\mathcal{G},I})\,P_{\mathcal{G},I}(a/b)$ is a nonzero integer. □

Remark 3.5. When the $m\times m$ block $M[X',Y']$ is invertible, the Schur complement gives $P_{\mathcal{G}}(x)=\det M[X',Y']\cdot\det(xI_C+\widetilde{M})$ with $\widetilde{M}=M[C,C]-M[C,Y']M[X',Y']^{-1}M[X',C]$. Thus the rectangular determinant is, up to a constant, the characteristic polynomial of a rational matrix of size $|C|$, as in the square case. We do not need this factorisation: Lemma 3.4 only uses the degree bound.

4. Arithmetic: the content

In this section we bound the common denominator of the coefficients of the members $P_{r,I}$ for the frames F_s. The method is that of Rhin and Viola [34, 35]: the p-adic valuation of the denominator is bounded prime by prime, and the bounds are summed against $\log p$ with the prime number theorem in arithmetic progressions. What is new is the form of the local bounds. Each of them is proved by its own argument, but all of them are explicit integer expressions in four integers attached to the prime, and their minimum, integrated, gives the content constant. The comparison between the local bounds and a table of affine pieces is a finite computation with rational numbers, and one theorem (Theorem 4.14) turns a table that passes it into the content estimate. The case of $L(2,\chi_{-3})$ is treated in Section 7.

4.1. **The root class of a prime.** Fix the frame F_s, with s of denominator q, and the alternating geometry of parameter $c = 2\alpha$. Put $N = r + 2m$. For a prime $p > q$ the indices $k \geq 1$ at which the entries of M acquire a denominator divisible by p are the multiples of p and the solutions of $qk \equiv o \pmod p$. Let $k = k_p$ be the least of the latter; then

$$qk_p = jp + o, \qquad 0 < j < q, \qquad j \equiv qs\,p^{-1} \pmod q.$$

We call j the *root class* of p and put $\kappa = \kappa_j = j/q$. Then $k_p - \kappa p = o/q \in [\frac{1}{q}, 1 - \frac{1}{q}]$: the first pole is at distance less than one from κp, uniformly in s. Throughout, $q(r+2m) < p^2$, so that every pole below N is simple and only the first power of p matters; the primes with $p^2 \leq q(r+2m)$ and the prime powers contribute $O(r^{3/2}\log r)$ in total.

4.2. **Gauss valuations, the assignment bound and the rank of the reduction.** Fix a prime p. Let $\mathcal{O} \subset \mathbb{Q}(x)$ be the localisation of $\mathbb{Z}_{(p)}[x]$ at the prime ideal (p); it is a discrete valuation ring with uniformiser p and residue field $\mathbb{F}_p(x)$, and its valuation v restricted to $\mathbb{Q}[x]$ is the *Gauss valuation* $v(f) = \min_k v_p(f_k)$, $f = \sum f_k x^k$. By Gauss's lemma $v(fg) = v(f) + v(g)$, and $v_p(\operatorname{den} f) = \max(0, -v(f))$. For a member we write $e_p(r,I) = v_p(\operatorname{den} P_{r,I})$.

Lemma 4.1 (assignment bound and LP duality)**.** *Let B be a square matrix over $\mathbb{Q}[x]$ with rows I and columns J, let $W_{tt'} = \max(0, -v(B_{tt'}))$, and let $\tau(W)$ be the largest weight $\sum W_{tt'}$ of a perfect matching between I and J. Then $v(\det B) \geq -\tau(W)$. If real numbers u_t, $v_{t'}$ satisfy $W_{tt'} \leq u_t + v_{t'}$ for all $t \in I$ and $t' \in J$, then $\tau(W) \leq \sum_t u_t + \sum_{t'} v_{t'}$.*

Proof. Each Leibniz term is a product of entries along a perfect matching, so its valuation is at least minus the weight of the matching, and v of a sum is at least the minimum. The second statement is weak LP duality for the assignment problem: the weight of a perfect matching is at most $\sum_t u_t + \sum_{t'} v_{t'}$. □

Lemma 4.2 (rank of the reduction)**.** *Let (u, v) be integer potentials with $W_{tt'} \le u_t + v_{t'}$ for the matrix $\Phi_{r,I}$ at p, of value $\tau' = \sum_t u_t + \sum_{t'} v_{t'}$. Then $B = \operatorname{diag}(p^u)\,\Phi_{r,I}\,\operatorname{diag}(p^v)$ has entries in $\mathcal{O}$, and*

$$e_p(r, I) \le \tau' - \operatorname{corank}_{\mathbb{F}_p(x)}(B \bmod p).$$

Proof. $v(\det \Phi_{r,I}) = -\tau' + v(\det B)$, and over the discrete valuation ring $\mathcal{O}$ the Smith normal form shows $v(\det B) \ge \operatorname{corank}(B \bmod p)$. □

The rank bounds below use a refinement of Lemma 4.2 at second order. Suppose that after a unimodular change of rows and columns over $\mathcal{O}$ the first n_1 rows and the first n_2 columns of B are divisible by p, and the $n_1 \times n_2$ block where they meet is divisible by p^2. Dividing the first n_1 rows by p leaves the first n_2 columns divisible by p, so

$$v(\det B) \ge n_1 + n_2. \tag{4.1}$$

The vectors that produce such rows and columns are kernel vectors of the reduction of B on single residue classes, and the condition on the corner is a statement about their pairing at first order, which we now describe. Let $B = p^2\Phi_{r,I}$ (all potentials equal to 1). Since $q(r + 2m) < p^2$, two distinct nodes $t \ne t'$ have $v_p(t'-t) \ge 1$ only if $t \equiv t' \pmod p$, and then $v_p(t'-t) = 1$. Hence B has entries in $\mathcal{O}$, its reduction is block diagonal over the residue classes modulo p (the *fibres*), and for nodes t, t' in distinct fibres

$$B_{tt'} \equiv \frac{p\,(V(t') - V(t))}{t' - t} \pmod{p^2}, \tag{4.2}$$

where $V(t) \in \mathbb{F}_p$ is the residue of p times the partial sum $\sum_{k \le t}(\frac{1}{k} - \frac{q}{qk-o})$, up to the normalising factor q^{-2}; V is a step function of t which jumps only at the poles $t \equiv 0$ and $t \equiv k_p$. Since $t' - t$ is prime to p, only its residue matters in (4.2).

For a vector $\mathbf{y}$ indexed by nodes, its *moment pair* is

$$\hat{\mathbf{y}} = \Big(\sum_t y_t,\ \sum_t y_t V(t)\Big).$$

Lemma 4.3 (pairing)**.** *Let $\mathbf{y}$ be supported on the rows of one fibre and $\mathbf{y}'$ on the columns of a different fibre, and let t and t' be nodes of the first and of the second fibre. Then*

$$\mathbf{y}^\top B\,\mathbf{y}' \equiv \frac{p}{t' - t}\,\det\begin{pmatrix}\hat{\mathbf{y}}\\ \hat{\mathbf{y}}'\end{pmatrix} \pmod{p^2}.$$

In particular the pairing vanishes modulo p^2 whenever the moment pairs of $\mathbf{y}$ and $\mathbf{y}'$ are proportional.

Proof. Expand $\mathbf{y}^\top B\,\mathbf{y}'$ with (4.2). □

The moment pairs are never both zero, by the displacement structure of Löwner matrices.

Lemma 4.4 (moment lemma)**.** *Let A be a matrix with rows indexed by a set R_0 of distinct integers and columns by a set C_0, $|R_0| = |C_0| + 1$, whose off-diagonal entries are $A(t,t') = (L(t') - L(t))/(t' - t)$ for some function L, and whose diagonal entries (on $R_0 \cap C_0$) are arbitrary. Then A has a nonzero left kernel vector $\mathbf{y}$ with $(\sum_t y_t, \sum_t y_t L(t)) \neq (0,0)$.*

Proof. With D_R, D_C the diagonal matrices of the indices, $D_R A - A D_C = L_R \mathbf{1}^\top - \mathbf{1} L_C^\top$ (the diagonal entries cancel on common indices). The left kernel $\mathcal{N}$ is nonzero. If every $\mathbf{y} \in \mathcal{N}$ had zero moment pair, multiplying on the left by $\mathbf{y}^\top$ would give $\mathbf{y}^\top D_R A = 0$, so D_R would preserve $\mathcal{N}$; as D_R is diagonal with distinct entries, $\mathcal{N}$ would contain a standard basis vector, supported on one index t, whose moment pair is $(1, L(t)) \neq (0,0)$. □

4.3. **The local bounds.** We now list the local bounds. Each is a statement of the following form: for every member $P_{r,I}$ with $|I| \leq K$, every prime $p > p_0$ with $q(r+2m) < p^2$ and every p satisfying the stated conditions,

$$e_p(r, I) \leq \tau(r, m, p, k_p) + \text{debit} \cdot |I|, \tag{4.3}$$

where τ is the stated expression and the *debit*, 0 or 1, is the stated cost of each deleted node. Here p_0 depends only on the frame. We write $(\cdot)_+ = \max(\cdot, 0)$, $\|m\|_p$ for the distance of m to $p\mathbb{Z}$, and $u_0 = (k_p + p - r)_+$.

Theorem 4.5 (local bounds)**.** *The following bounds of the form (4.3) hold.*

	bound τ	*conditions*	*debit*
(T1)	$2(r+m)$	*none*	0
(T1′)	$2r + 2m - \|m\|_p$	*none*	0
(T2)	$2(r-u_0)_+ + 2m + \min(u_0, 2m)$	$2m < p \leq r$	0
(T3)	$2r + m$	$2m < p \leq r$	0
(T4)	$2(r-k_p)_+ + 2m + (2m - (p-r))_+$	$r < p$	0
(T5)	$(2m - (k_p - r)_+)_+$	$N \leq p$, $r \leq k_p$	0
(LP1)	$\tau_{\mathrm{T2}} - (\min(u_0, 2m) - m)_+$	$p \leq r < p + k_p$, $N \leq 2p$	1
(LP2)	$2(r - k_p) + 3m$	$r < p < r + m$, $k_p \leq r$, $N \leq 2p$	1
(W)	$2r + 2m - (t_2 - t_1)$ *for each window* $[t_1, t_2)$ *of Theorem 4.7*	*as there*	0
(R)	Π_R *of* (4.4)	$2m < p$	0
(S)	Π_S *of* (4.5)	$2m \leq p$	1
(U_n)	$\Pi_{U,n}$ *of* (4.6), $n \geq 0$	$np \leq m < (n+1)p$, *and* $p \leq 2m$ *if* $n = 0$	0

The rest of this subsection proves Theorem 4.5, by family.

4.3.1. *The assignment bounds (T1)–(T5).* These follow from Lemma 4.1 with explicit potentials. By the ultrametric inequality, an off-diagonal entry $M[t,t']$ has $-v_p \le 1$ if the interval $(t,t']$ contains a pole, plus 1 if $p \mid t'-t$, and 0 otherwise; a diagonal entry $M[t,t]$ has $-v_p \le 2$ if $t \ge k_p$ and 0 if $t < k_p$. (T1) is the trivial bound with potentials 1 on every row and column. For (T1′) the potentials are $\frac{1}{2}, \frac{3}{2}, 1$ on rows and $\frac{3}{2}, \frac{1}{2}, 1$ on columns of a residue class modulo p, according as X' has more, fewer or as many nodes in the class as Y'; since the border alternates between X' and Y', the imbalance is at most one per class and the number of unbalanced classes is $2\|m\|_p$. For (T2)–(T5) one starts from the potential $\mathbf{1}[t \ge k_p]$ on the common nodes t, which pays for the diagonal, and adds 1, or $1 + \mathbf{1}[z \ge p]$, on the border nodes z, raising the potentials of the common nodes that have a congruent partner in the border. In each range one checks $W \le u + v$ entrywise, using two facts: two distinct nodes not congruent modulo p have weight at most 1, and two border nodes are never congruent once $2m < p$. For (T5), with $N \le p$ and $r \le k_p$, no common node pays for the diagonal and only the border nodes beyond k_p carry potential 1; in particular (T5) vanishes once $k_p \ge N$, which happens for $p/r > \max(1 + 2\alpha, 1/\kappa) + o(1)$. All cases are checked in the formal proof.

4.3.2. *Primes near r: the bounds (LP1) and (LP2).* For p close to r the reduction of the rescaled matrix has a large block of rank one per residue class, and the corank can be computed. We give the proof of (LP2). The bound (LP1), for the primes with $p \le r < p + k_p$ and $N \le 2p$, is proved by the same method: with the potentials of (T2), the elimination of an invertible block of common nodes brings the reduction to the form $\left(\begin{smallmatrix} \tilde{x}I & G_Y \\ G_X & 0 \end{smallmatrix}\right)$, with $\tilde{x}$ the variable x plus a constant, in which the border nodes below $p + k_p$ contribute rows and columns proportional to unit vectors, and the Frobenius–König theorem bounds its rank. The saving over (T2) is $(\min(2m, p + k_p - r) - m)_+$, where $\min(2m, p + k_p - r)$ is the number of border nodes below $p + k_p$; we refer to the formal proof for the details.

Theorem 4.6. *Assume $r < p < r + m$, $k = k_p \le r$ and $N = r + 2m \le 2p$, and that every pole below N is simple. Then $v(\det \Phi_{r,\emptyset}) \ge -(2(r-k) + 3m)$. The same holds for every member, with debit 1 per deleted node.*

Proof. Use the potentials $u(t) = \mathbf{1}[k \le t]$ for common nodes $t < r$ and $u(t) = 1 + \mathbf{1}[p \le t]$ for border nodes t, of value $\tau' = 2(r-k) + 2m + (N-p)$. Since $N \le 2p$, the poles below N are k, p, and $p + k$ if $p + k < N$; each is simple, and the residue of p times the increment $\frac{1}{t} - \frac{q}{qt-o}$ at them is $-q/j$, 1 and $-q/(q+j)$ respectively.

Rows and columns beyond $p+k$ vanish. Let $z \in [p+k, N)$ be a border node, so $u(z) = 2$, and let t be a node of the other kind. If $t < k$, then $u(t) = 0$, the partial sum over $(t, z]$ has valuation at least -1 and $z - t \in (p, 2p)$ is prime to p; otherwise $u(t) \ge 1$ and $v_p(M[t,z]) \ge -2$. In both cases the rescaled entry is divisible by p.

The remaining entries. For nodes below $p+k$ the only poles are k and p. With $B_0=[0,k)$ and $B_1=[k,r)$ the reduction has the following blocks, with $\omega=q^{-2}$ the normalisation of (2.1): on $B_0\times B_0$ the regular part plus x on the diagonal; on $B_0\times B_1$ the entries $-\omega q/(j(t-t'))$ ($t'\in B_0$, $t\in B_1$); on $B_1\times B_1$ the scalar matrix $\omega(q/j)^2 I$; a border node $z\in[r,p)$ against $t'\in B_0$ gives $\omega q/(j(z-t'))$; a border node $z\in[p,p+k)$ against $t'\in B_0$ gives $\omega(1-q/j)$ if $t'=z-p$ and 0 otherwise; all other border entries vanish.

Rank. Delete the zero rows and columns found above; this does not change the rank. Eliminating the scalar block on B_1, the Schur complement on B_0 is $\tilde{x}I$ plus a constant multiple of the matrix with entries $\sum_{i=1}^{p-r}\frac{1}{(t+i)(t'+i)}$ ($t,t'\in B_0$), where $\tilde{x}$ is x plus a constant, by the reflection $H_{p-1-n}\equiv H_n \pmod p$ of harmonic numbers. The border nodes $z\in[r,p)$, indexed by $i=p-z$, are all present because $N>p$; even offsets are rows and odd offsets are columns, and together they cover $1\le i\le p-r$, so row and column operations remove that multiple and leave $\left(\begin{smallmatrix}\tilde{x}I & G_Y\\ G_X & 0\end{smallmatrix}\right)$. A Schur complement on $\tilde{x}I$ (here it is essential that x is an indeterminate, so that $\tilde{x}$ is a unit) shows that its rank is $k+\operatorname{rank}(G_XG_Y)$. In G_XG_Y the block between the high rows and the high columns is zero, since their offsets $z-p$ have opposite parities, so $\operatorname{rank}(G_XG_Y)\le p-r$. In total the rank of the reduction is at most $(r-k)+k+(p-r)=p$, the corank is at least $r+m-p$, and Lemma 4.2 gives $v(\det)\ge(r+m-p)-\tau'=-(2(r-k)+3m)$. Deleting a common node changes the potentials by at most one. □

Under the hypothesis $N\le p+k$ the third pole does not occur and the proof is shorter; the observation that the border nodes beyond $p+k$ only contribute rows and columns that vanish modulo p allows the whole range $N\le 2p$, which on the interval $r<p<r+m$ is automatic.

4.3.3. *Pole-free windows: the bound (W).* Like the rank bounds, the bound (W) does not come from the valuations of the entries alone. Between two consecutive poles the step function V of (4.2) is constant, and the reductions of the fibres of nearby residue classes are then translates of one another. Kernel vectors can be transported from one fibre to the next, and their pairings vanish modulo p^2 by Lemma 4.3; the corner bound (4.1) turns this into a saving at second order.

A *cut* is an integer t with $t\equiv 0$, $t\equiv k_p$ or $t\equiv r \pmod p$. A window $[t_1,t_2)$ of integers is *cut-free* if no cut lies in the open interval (t_1,t_2).

Theorem 4.7. *Assume that every pole below N is simple and $N<p^2$. Let $n\ge 0$ with $np\le m$, put $N_n=r+2(m-np)$, and let $[t_1,t_2)$ be a cut-free window with $\max(r,N_n-p)\le t_1\le t_2\le\min(N_n,r+p)$. Then every member satisfies*

$$e_p(r,I)\le 2r+2m-(t_2-t_1).$$

For $n=0$ the windows lie in the border above $\max(r,N-p)$, and the bound applies to primes $p>m$. For $n=\lfloor m/p\rfloor\ge 1$ it applies to the small

primes $p \le m$, where among the assignment bounds only (T1) and (T1′) are available. Read modulo p, the window range is then the arc of residues that receive an odd number of border nodes, and the saving over (T1′) is the amount, if any, by which the length of the longest cut-free run of this arc exceeds $\|m\|_p$. For other n the window range is empty. The proof uses the frame only through its poles: for a frame with several digamma terms the same statement holds when the cuts are the residues of all the poles and of r.

Proof. Let $B = p^2\Phi$, with independent diagonal variables on the common nodes (every member is a coefficient of this determinant, so a bound for it bounds every member). Let $i < i'$ be two positions of the window; they lie in different residue classes. Translation $t \mapsto t + (i' - i)$ maps the nodes of the fibre of i bijectively onto those of the fibre of i': common nodes to common nodes, since no $t \equiv r$ lies in $(i, i']$; border nodes to border nodes, since both classes lie in the window range; and X' to Y' when $i' - i$ is odd. Since there is no pole in $(i, i']$, the step function V and the diagonal residues are unchanged, and so are the differences of nodes. Hence the reduced block of the fibre of i' equals that of i when $i' - i$ is even, and its transpose when $i' - i$ is odd. In the window the fibres alternate between one more row than column and one more column than row. A nonzero left kernel vector $\mathbf{y}$ of one block, transported to every fibre of the window, is a left kernel vector on the fibres with a surplus row and a right kernel vector on the others, with the same moment pair everywhere. By Lemma 4.3 all pairings between these vectors vanish modulo p^2. Extending the n_1 vectors on rows and the n_2 vectors on columns, $n_1 + n_2 = t_2 - t_1$, to unimodular matrices, (4.1) gives $v(\det B) \ge t_2 - t_1$. Since $\det B = p^{2(r+m)} \det \Phi$, the theorem follows. □

4.3.4. *The rank bound (R).* For $2m < p$ write $r = hp + r_p$ with $0 \le r_p < p$ (note that $k_p = \lceil jp/q \rceil$, since $0 < o < q$), and put $\mathrm{arc}_0 = [0, k_p - 1]$, $\mathrm{arc}_1 = [k_p, p - 1]$. For $k \ge 0$ and $z \in \{0, 1\}$ let

$$n_{k,z} = \big|[r, r + 2m) \cap [kp + \min \mathrm{arc}_z,\ kp + \max \mathrm{arc}_z]\big|,$$
$$n_{\mathrm{sq}} = \big|[\max(0, r + 2m - p), \min(r, k_p))\big|,$$

and

(4.4)
$$\Pi_R = 2(r+m) - \max\big(m + n_{\mathrm{sq}},\ n_{0,0} + n_{1,0} + 2n_{\mathrm{sq}},\ n_{h,0},\ n_{h,1},\ n_{h+1,0},\ n_{h+1,1}\big).$$

Proof of the bound (R). Every node is $t = i + pk$ with residue $i \in [0, p)$ and digit k. Write $k(i)$ for the number of common nodes in the fibre of i. Since the border has length $2m < p$, this fibre contains the common digits $0, \dots, k(i) - 1$ and at most one border digit, namely $k(i) \in \{h, h + 1\}$, which is a row or a column according to the parity of $i + pk(i) - r$; the step function V is constant on the fibre up to the bit $z(i) = \mathbf{1}[i \ge k_p]$. Thus the reduced block of a fibre depends only on $(k(i), z(i))$ and the role of its border digit, and its entries have the Löwner form of Lemma 4.4. A fibre with a border

row (column) has a left (right) kernel vector with nonzero moment pair; for the digit 0 below the pole the step function vanishes, so these fibres share the moment direction $(1,0)$, and a fibre of type $(1,0)$ without border digit has the zero block and contributes one row and one column. Choosing one such vector per fibre kills $m + n_{\mathrm{sq}}$ rows and as many columns, which gives the saving $m + n_{\mathrm{sq}}$. Within a class of fibres of the same type (k,z) all moment pairs are equal, so by Lemma 4.3 the pairings between the killed rows and columns of one class vanish modulo p^2, and (4.1) gives a saving equal to the number of killed coordinates of the class: $n_{k,z}$ for $k \geq 1$ and $(k,z) \neq (1,0)$, and $n_{0,0} + n_{1,0} + 2n_{\mathrm{sq}}$ for the class of direction $(1,0)$. Taking the best class gives (4.4). A deleted common node changes at most one fibre; giving it, and one further fibre to restore the balance of rows and columns, the trivial certificate costs at most two killed coordinates of each kind, which the smaller matrix compensates, so the debit is 0. □

4.3.5. *Separated border nodes: the bound (S).* The last bound uses a different source of cancellation. Choose a window $[t_0, t_v]$ of border nodes whose residues lie in one arc between consecutive poles, and consider a border node outside it. If the interval between the node and t_0 contains a singular index, the residues modulo p of p times the partial sums at the two nodes differ (two singular contributions in an interval of length less than p cannot cancel). The leading coefficient in x of the bordered determinant of a fibre is this residue, so the cross product of the two bordered fibre determinants has a nonzero leading coefficient, and the border node is eliminated against t_0 with no further information. Counting the residues of the window together with the residues of the eliminated border nodes gives an explicit bound, a minimum over the four window types $(k,z) \in \{h, h+1\} \times \{0,1\}$, which we now state.

Let $2m \leq p$, and let h, r_p, the arcs arc_0, arc_1 and the counts $n_{k,z}$ be as in §4.3.4. For a type (k,z) let $\mathcal{K}_{k,z}$ be the interval of the integers $kp + i$ with $i \in \mathrm{arc}_z$, and with $i < r_p$ if $k = h+1$; the border nodes in $\mathcal{K}_{k,z}$ form the window, and there are $n_{k,z}$ of them. Let n_K be the number of border nodes in $\mathcal{K}_{k,z}$ and in its two translates $\mathcal{K}_{k,z} \pm p$, and put

$$n_\emptyset = p - |\mathcal{K}_{k,z}| - (2m - n_K), \qquad n_+ = (n_{k,z} - m)_+.$$

Since $2m \leq p$, each residue class receives at most one border node, so n_K is the number of border nodes whose residues are those of $\mathcal{K}_{k,z}$, and $n_\emptyset$ is the number of the other residue classes that receive no border node. Then

$$\Pi_S = \min_{(k,z)} \bigl(2r + m - n_+ - (n_+ - n_\emptyset)_+\bigr), \tag{4.5}$$

where a type whose window is empty contributes $2r + m$, and (S) is the bound $e_p(r, I) \leq \Pi_S + |I|$. The proof carries out the elimination above for every border node outside the window; we refer to the formal proof for it.

4.3.6. *The border wrapping around: the bound (U_n).* When $p \le 2m$ the border meets some residue classes more than once, and the bound (S) is not available. The last bound extends it to these primes. It works at third order in the Smith normal form: we show that, after the elimination of a block of unit pivots, the rescaled matrix has the Smith invariants of a smaller geometry to which the separation argument of (S) applies, and that the third-order information of that argument survives the elimination.

Write $m = np + e$ with $0 \le e < p$, so that $n = \lfloor m/p \rfloor$, and put

$$d = \|m\|_p = \min(e, p-e), \qquad a = r + m - d.$$

For $j \in \{0,1,2\}$ and $z \in \{0,1\}$ let $B_j = p\lfloor r/p \rfloor + (n+j)p$, let $[\lambda_0, \upsilon_0] = [0, k_p - 1]$ and $[\lambda_1, \upsilon_1] = [k_p, p-1]$ be the two arcs between the poles, and put

$$w_0 = B_j + \lambda_z, \qquad t_0 = \max(a, w_0), \qquad t_v = \min(B_j + \upsilon_z,\, a + p - 1).$$

In the virtual geometry with common block $[0, a)$ and border $[a, a+2d)$, let $n_+ = \big(\min(t_v+1, a+2d) - t_0 - d\big)_+$, let n_K be the number of border nodes in $[w_0, t_v]$ and its two translates by $\pm p$, and let $n_\emptyset = p - (t_v + 1 - w_0) - (2d - n_K)$. Then

$$\Pi_{U,n} = \min_{j,z} \Big(2(r+m) - d - n_+ - (n_+ - n_\emptyset)_+ \Big), \tag{4.6}$$

all subtractions being truncated. This is the expression (4.5) for the virtual geometry (a, d), with the windows $[w_0, t_v]$ in place of $\mathcal{K}_{k,z}$; note that $2(r+m) - d = 2a + d$.

Theorem 4.8. *Let $p > p_0$ be a prime with $q(r+2m) < p^2$, let $n \ge 0$ with $np \le m < (n+1)p$, and assume $p \le 2m$ if $n = 0$. Then every member satisfies $e_p(r, I) \le \Pi_{U,n}$.*

We first describe the proof for $n = 0$, that is $m < p \le 2m$, and then the changes for $n \ge 1$. As in the proof of Theorem 4.7 we work with $B = p^2\Phi$ and an independent indeterminate U on each common diagonal entry, over the discrete valuation ring $\mathcal{O}$ of Gauss valuations in these indeterminates; every member is a coefficient of $\det B$, up to the deletion of common rows and columns, which we treat at the end.

Balanced pivots. For $n = 0$ there are $b = 2m - p$ residue classes carrying two border nodes $\ell < \ell + p$, one a row and the other a column, and $2d$ classes carrying one border node; the latter lie in an interval of length $2d < p$. Eliminate the common block, whose fibres are invertible because U is an indeterminate. The remaining 2×2 fibre of a class with two border nodes has the pivot $\delta = h - s_X^\top Q^{-1} s_Y$, whose limit as $U \to \infty$ is a nonzero multiple of the residue of p times the increment of the frame over $(\ell, \ell + p]$. This interval contains exactly one pole of each digamma term. If the two contributions cancelled, the pole t of the second term would satisfy $q(t - \ell p) \equiv o$ with $|q(t - \ell p) - o| < q$, hence $q(t - \ell p) = o$, which is impossible for $0 < o < q$. So these b pivots are units. After them, the matrix of size $r + m = a + d$ is

equivalent over $\mathcal{O}$ to $\operatorname{diag}(I_a, pH)$ with H of size d, and its Schur complement on the border has the expansion

$$S = p\,S_1 + p^2(S_2 - C_{UB}D_B^{-1}C_{BU}) + O(p^3), \tag{4.7}$$

where D_B is the diagonal of the balanced pivots and C_{UB}, C_{BU} are the first-order couplings between them and the single border nodes. The correction term in (4.7) is not negligible; it is accounted for by the following identities.

Lemma 4.9 (bordered inverses)**.** *Let Q be an invertible square matrix over a commutative ring and u a vector.*

(a) If M is obtained from Q by replacing one column by a vector v, M is invertible, and u' is obtained from u by replacing the corresponding entry by $u^\top Q^{-1}v$, then $u'^\top M^{-1}u = u^\top Q^{-1}u$.

(b) If the matrix $M = \left(\begin{smallmatrix} Q & s_Y \\ s_X^\top & h \end{smallmatrix}\right)$ is invertible and $z_X = s_X^\top Q^{-1}u$, then $(u, z_Y)^\top M^{-1}(u, z_X) = u^\top Q^{-1}u$ for every z_Y.

No symmetry of Q or M is assumed.

Proof. (a) Comparing columns gives $u'^\top = u^\top Q^{-1}M$. (b) $M\,(Q^{-1}u, 0)^\top = (u, s_X^\top Q^{-1}u)^\top = (u, z_X)^\top$. □

Moments. In the separation argument of (S), a border node is credited when its moment relative to a chosen reference node vanishes, and that moment is a bilinear form $u^\top Q^{-1}u$ in the inverse of the common fibre. After the elimination, the common fibre of a class with two border nodes is replaced by a mixed block M as in Lemma 4.9, with the lower node as a row and the upper node as a column, or conversely. The added product in the inverse of M is exactly the product subtracted in (4.7), and Lemma 4.9 shows that the mixed moment equals the old one on every class that the separation argument credits. The single border nodes outside the window are separated from it as in (S), because two singular contributions in an interval of length $2d < p$ cannot cancel. The count of (S), applied to the virtual geometry (a, d), then gives (4.6) through the third-order analogue of (4.1): if $\sigma = n_+$ kernel vectors on each side survive and at most ς residue classes carry a nonzero source, then $\sum_i \min(v_i, 3) \geq d + \sigma + (\sigma - \varsigma)_+$, where the v_i are the valuations of the Smith invariants and the first d is the factor p of pH.

Members. Deleting k common nodes lowers the trivial bound $2(r + m)$ by $2k$. A class with one deleted common node and an unchanged direction keeps its moment: if Q is the common fibre and the last coordinate is deleted, the residual of the smaller solution is supported on that coordinate, and the equality of the two directions forces its pairing with the load to vanish. A class whose direction changes gives a new separation instead. Classes with two or more deleted nodes, and classes outside the window, cost at most one each. With t_1, t_2 and t' the numbers of classes of these three kinds, $t_1 + 2t_2 + t' \leq k$, and an elementary inequality for $f(\sigma, \varsigma) = \sigma + (\sigma - \varsigma)_+$

shows that the loss is at most $2k$. Hence the bound holds for every member with debit 0.

Larger n. For $np \le m < (n+1)p$ each class of the remaining border absorbs n pairs of border nodes, and its block after the elimination is a mixed block whose rows and columns are the nodes of $2n+1$ consecutive quotients $\beta, \beta+1, \dots, \beta+2n$, where β is the number of common nodes in the class (the quotient of a node t in the class of γ is $(t-\gamma)/p$). The window counts are those of the first case at the scale $r+np$, translated by np. The new difficulty is to show that a remaining border node separated from the window by a pole really has a nonzero coupling, because the directions are now those of $(n+1)\times(n+1)$ bordered systems.

Lemma 4.10 (bordered systems and rational interpolation)**.** *Let F be a field, $S = \{0,1,\dots,2n\}$ with $(2n)! \neq 0$ in F, $L : S \to F$, and $S = X \sqcup Y$ with $|X| = n+1$, $|Y| = n$. The solutions (v,δ) of the bordered system*

$$\sum_{y\in Y} \frac{L(y)-L(x)}{y-x}\, v_y + \delta = L(x) \qquad (x \in X)$$

correspond bijectively to the pairs (N,D) of polynomials with D monic of degree n, $\deg N \le n$ and $N(t) = L(t)D(t)$ for $t \in S$, with δ the leading coefficient of N. In particular the system is nonsingular exactly when the interpolation problem of type (n,n) has a unique solution, and then $\delta = \lim_{t\to\infty} N(t)/D(t)$ depends only on L and not on the split.

Proof. Let $W(t) = \prod_{y\in Y}(t-y)$. Given (v,δ) put $D = W - \sum_y v_y W'(y)\,\omega_y$ and $N = \delta W - \sum_y v_y L(y) W'(y)\,\omega_y$, where $\omega_y = W/((t-y)W'(y))$ are the Lagrange polynomials of Y. Then $N(y) = L(y)D(y)$ for $y \in Y$, and dividing $N(x) - L(x)D(x)$ by $W(x) \neq 0$ gives the equation of the row x. Conversely, Lagrange interpolation on Y recovers $v_y = -D(y)/W'(y)$ and δ from (N,D). □

Lemma 4.11 (harmonic loads)**.** *Let $A \in F$ with $A + h \neq 0$ for $0 \le h < 2n$, let $\ell_A(s) = \sum_{0\le h<s} 1/(A+h)$, and let the characteristic of F be 0 or larger than $2n$. For every split, the bordered system of Lemma 4.10 with $L = \ell_A$ is nonsingular, and its direction is $\delta = \ell_A(n) + c_n$ with $c_n \in F$ depending only on n. Consequently, for the harmonic load $L(t) = H_t$ on the quotients $\beta,\dots,\beta+2n$ the direction is $H_{\beta+n} + c_n$, and the directions of two classes with β and $\beta+1$ common nodes differ by $1/(\beta+n+1)$.*

Proof. Write Δ for the forward difference. One has $\Delta^j \ell_A(k) = (-1)^{j-1}(j-1)!/(A+k)_j$ for $j \ge 1$, with $(x)_j$ the rising factorial, and $\Delta^m[\binom{s}{k} f(s)](0) = \binom{m}{k}\Delta^{m-k} f(k)$. By Lemma 4.10 a solution is a pair (N,D) with $N = \ell_A D$ on S, that is, with $\Delta^m(\ell_A D)(0) = 0$ for $n < m \le 2n$, because N has degree at most n. Writing $D = \sum_k d_k \binom{t}{k}$ and inserting the two formulas, these n equations become, after removing nonzero factors of the rows and columns, a system with the Cauchy matrix $\big(1/(m-k)\big)_{n<m\le 2n,\, 0\le k<n}$ for the coefficients d_k, $k < n$, with a right-hand side coming from $d_n = 1$. The

Cauchy matrix is nonsingular, so D is unique, which is the nonsingularity. The same computation shows that $d_k/(A+k)_{n-k}$ does not depend on A, and then, since $\Delta^{n-k}\ell_A(k)$ is a constant multiple of $1/(A+k)_{n-k}$ for $k<n$, $\delta = \Delta^n(\ell_A D)(0)/n! = \ell_A(n)+c_n$ with c_n independent of A. For the harmonic load on the quotients $\beta+s$ we have $H_{\beta+s} = H_\beta + \ell_{\beta+1}(s)$; a constant added to L is added to δ, so the direction is $H_\beta + \ell_{\beta+1}(n) + c_n = H_{\beta+n} + c_n$. □

Remark 4.12. The interpolation problem of Lemma 4.11 can be solved in closed form. Its denominator is the polynomial $D_n(t) = \sum_{k=0}^{n} \binom{n}{k}\binom{2n-k}{n-k}(A+k)_{n-k}\, t(t-1)\cdots(t-k+1)$, of Hahn type [19], the constant is $c_n = 2H_n$, and the determinant of the bordered system is $\mathrm{sf}(n)^2\,\mathrm{sf}(n-1)^2/(\prod_{x\in X, y\in Y}|x-y|\cdot\prod_{i<n}(A+i)_{n+1})$, with $\mathrm{sf}(n) = \prod_{i\le n} i!$; all of its prime factors are at most $2n$ or divide some $A+h$. These closed forms are not used in the proof and are not part of the formalization.

Separation for $n \ge 1$. Between a remaining border node and the reference node of the window lies a cut. If it is a pole of the second digamma term, deform the shift to $s+pY$ with an indeterminate Y and evaluate at the value of Y at which that pole becomes central: after multiplication by the vanishing factor, the bordered systems of the two classes become block systems built from Cauchy matrices on the quotients, the square blocks are nonsingular by Cauchy's determinant formula, and a rectangular Cauchy matrix with $k+1$ rows and k columns has a left kernel vector with nonzero coordinate sum (otherwise $\sum_j b_j/(u_j-T)$ would have k roots and a numerator of degree less than k). This gives the two directions the values 0 and the weight of the term, which differ. If the cut is a multiple of p, take instead the leading coefficient in Y: every fractional pole term tends to 0, the loads become the harmonic numbers, the two classes have β and $\beta+1$ common nodes, and Lemma 4.11 gives directions differing by $1/(\beta+n+1)$, a unit because $\beta+2n+1<p$ by $q(r+2m)<p^2$. In both cases the comparison is made on the cleared numerators of the original determinants, not on inverses, and the result returns to $Y=0$ by Gauss's lemma: the cleared denominators are products of the elementary denominators of the entries, which are units at $Y=0$, so a lower bound for the Gauss valuation of the deformed determinant gives the same lower bound for every coefficient of the original one. With these separations, the count and the treatment of the members are those of the case $n=0$.

4.4. **Profiles.** Let $\mathcal{E}$ be the set of integer expressions in r, p, m, k built from integer constants with $+$, multiplication by integer constants, truncated subtraction $(X-Y)_+$, min, max, the distance $\|X\|_p$ of an expression X to $p\mathbb{Z}$, and the quantities $p\lfloor r/p\rfloor$ and $r \bmod p$. Every bound of Theorem 4.5 lies in $\mathcal{E}$; for (R) and (S) this is (4.4) and (4.5), the ends of the intervals $\mathcal{K}_{k,z}$ being $kp = p\lfloor r/p\rfloor + (k-h)p$ plus 0, k_p, k_p-1, $p-1$ or r_p-1, for (U_n) this is (4.6) with $d = \|m\|_p$ and $B_j = p\lfloor r/p\rfloor + (n+j)p$, and for (W) the window ends are the cuts nearest to the ends of the window range, which are expressions

of the same kind. The *homogenisation* $\hat{\tau}(\theta)$ of $\tau \in \mathcal{E}$ replaces r, p, m, k by $1, \theta, \alpha, \kappa\theta$, every constant by 0, $\|X\|_p$ by $\theta\|\hat{X}/\theta\|$ and $p\lfloor r/p \rfloor$ by $\theta\lfloor 1/\theta \rfloor$. It is piecewise affine with rational breakpoints, finitely many in every interval bounded away from 0.

Lemma 4.13 (homogenisation)**.** *For each $\tau \in \mathcal{E}$ there is an explicit constant $E(\tau)$, computed recursively from $E(m) = \frac{1}{2}$, $E(k) = 1 - \frac{1}{q}$ and the rules $E(\tau + \tau') = E(\tau) + E(\tau')$, $E(\min(\tau, \tau')) \le \max(E(\tau), E(\tau'))$, such that $|\tau(r, p, m(r), k_p) - r\hat{\tau}(p/r)| \le E(\tau)$ for all $r, p \ge 1$.*

Proof. The real version of τ, with the constants dropped, is positively homogeneous of degree one, and every operation is 1-Lipschitz in the maximum norm. The inputs differ from their homogeneous values by $|m - \alpha r| \le \frac{1}{2}$ and $|k_p - \kappa p| \le 1 - \frac{1}{q}$. Induct on τ. □

The conditions of Theorem 4.5 are linear inequalities $L \ge 0$ or $L > 0$ in r, p, m, k, and homogenise to intervals U of θ. At an endpoint the error $L - r\hat{L}(p/r)$ is a combination of $m - \alpha r$ and $k_p - \kappa p$. With the rounding $m = \lceil \alpha r - \frac{1}{2} \rceil$ the endpoints $\theta = 2\alpha$ (from $2m < p$) and $\theta = (1+2\alpha)/(1+\kappa)$ (from $N \le p + k$) are *exact*: the condition holds for every prime on the correct side. The other endpoints are exact or hold outside a strip of primes of bounded width around the endpoint, on which another bound must be used.

4.5. **The checker and the content theorem.** For the root class j let $\mathcal{P}_j(\theta)$ be the minimum of the homogenised bounds $\hat{\tau}_b(\theta)$ over the bounds b valid at θ. A *content table* for the frame F_s and the geometry of parameter c consists of the parameters (q, α, K) and, for every root class j, an increasing sequence $0 < \theta_0 < \theta_1 < \cdots < \theta_{\mathrm{end}}$ with, on each piece $(\theta_{i-1}, \theta_i]$, an affine function $\ell_i(\theta)$ and a constant $C_i \ge 0$, each piece tagged with a bound b of Theorem 4.5 and, at each inexact endpoint of the validity interval of b, a second bound valid across it; together with a constant ℓ_0 for the prefix $(0, \theta_0]$ and the rational number Λ_{fin}. The *checker* accepts the table if:

(C1) for each shift of the table, every unit class modulo the modulus M of the prime number theorem used is assigned a root class with a table;
(C2) on each piece ℓ_i is nonnegative, the piece lies in the validity interval of its bound, and $\hat{\tau}_b$ lies below ℓ_i at both ends and at both one-sided limits of each breakpoint inside the piece;
(C3) at each inexact endpoint the second bound is valid on both sides and lies below ℓ_i there, and C_i exceeds the error constants of Lemma 4.13, the debits times K, the strip widths times the slopes, and the trivial bound at the scales below an explicit threshold;
(C4) $\ell_0 \ge 2 + 2\alpha$, $0 < \theta_0 \le 1$, and $\theta_{\mathrm{end}} \ge \max(q(1 + 2\alpha), p_0/2)$ with $\kappa_j \theta_{\mathrm{end}} \ge 1 + 2\alpha$;
(C5) $\frac{1}{\varphi(q)} \sum_j (\ell_0\theta_0 + \sum_i \int_{\theta_{i-1}}^{\theta_i} \ell_i(\theta)\, d\theta) = \Lambda_{\mathrm{fin}}$.

All of these are comparisons of rational numbers.

Theorem 4.14 (content)**.** *Let $0 < c \le 1$, let F_s be a frame with s of denominator q, and suppose that the checker accepts a content table with s among its shifts and depth K. Then for every even r and every member with $|I| \le K$,*

$$\log \operatorname{den} P_{r,I} \le (\Lambda_{\mathrm{fin}} + o(1))\, r^2.$$

Proof. By definition $\operatorname{den} P_{r,I} = \prod_p p^{e_p(r,I)}$. Fix a piece $(\theta_{i-1}, \theta_i]$ of the root class j and a prime p in a unit class assigned to j with $\theta_{i-1} r < p \le \theta_i r$. For r beyond the threshold of (C3) the conditions of the tagged bound hold (exactly, or outside a strip where the second bound applies), and Lemma 4.13 with (C2), (C3) gives $e_p(r, I) \le r\,\ell_i(p/r) + C_i$; below the threshold the trivial bound is at most C_i. For $p \le \theta_0 r$ the bound (T1) gives $e_p \le \ell_0 r + O(1)$, with the primes $p^2 \le q(r + 2m)$, the prime powers and the finitely many primes below p_0 contributing $O(r^{3/2} \log r)$. For $p > \theta_{\mathrm{end}} r$, (C4) gives $N \le p$ and $k_p \ge N$, and (T5) gives $e_p = 0$. By the prime number theorem in arithmetic progressions modulo M, for each unit class $\sum_{\theta_{i-1} r < p \le \theta_i r,\ p \equiv \cdot} r\,\ell_i(p/r) \log p = \frac{r^2}{\varphi(M)} \int_{\theta_{i-1}}^{\theta_i} \ell_i(\theta)\, d\theta + o(r^2)$, and the class j receives $\varphi(M)/\varphi(q)$ unit classes modulo M, because the map sending a unit to qs times its inverse permutes the units modulo q. Summing and using (C5) gives the theorem; the constants C_i contribute $O(r)$. □

The tables used in this paper are generated from the functions $\mathcal{P}_j$ themselves: the pieces are the maximal intervals on which $\mathcal{P}_j$ is affine and attained by one bound, so that $\Lambda_{\mathrm{fin}} = \frac{1}{\varphi(q)} \sum_j \int \mathcal{P}_j$ up to the prefix. Every bound of Theorem 4.5 is available to the tables: (T1)–(T5), (LP1), (LP2), (R), (S), the windows (W) and the bounds (U_n). The windows with $n = 0$ are used for $m < p$, between consecutive cuts. On $(0, \alpha]$, that is for $p \le m$, the only other bounds are (T1), (T1′) and (U_n); with (T1′) alone the profile there is the sawtooth $2 + 2\alpha - \theta \| \alpha/\theta \|$, whose integral is

$$\int_0^{\alpha} \Big(2 + 2\alpha - \theta \Big\| \frac{\alpha}{\theta} \Big\| \Big) d\theta = 2\alpha + \frac{\alpha^2}{2} + 2\alpha^2 \log 2,$$

since $\int_1^{\infty} \|y\| y^{-3}\, dy = \frac{3}{2} - 2 \log 2$. The windows with $n = \lfloor m/p \rfloor \ge 1$ lower this profile on the strips $\alpha/(n+1) < \theta \le \alpha/n$; the tables use them for $n \le 8$, again with each window running between two consecutive cuts. For $q = 3, 4, 5, 7$ the tables also use (U_0) on $(\alpha, 2\alpha]$ and (U_n) on the strips with $1 \le n \le 24$; the table for $q = 2$ uses only the other bounds. Beyond the strips used, a table resolves the sawtooth into teeth down to α/n_0 and ends with a constant piece on $(0, \alpha/n_0]$, which costs $\alpha^2 T_{n_0}$ with $0 < T_n \le 1/(4n^2)$; the number Λ_{fin} is then rational, and (C5) is an identity between rational numbers. The bounds depend on the shift s only through the root class, and their profiles only through (q, j, α), so one table serves all shifts with denominator q.

5. Nonvanishing

The deduction needs, for every rational x_0 and every large scale, one member of the family that does not vanish at x_0. For the frames of (2.1) this holds in characteristic zero, with a number of deleted common nodes bounded independently of r; this is the statement used in Section 8.

5.1. **Selection in characteristic zero.** Normalise the frame so that the divided differences of F at consecutive nodes have prefix values f_t, with $f_{t+1}-f_t=\Delta(t)$, and the diagonal prefix e_t satisfies $e_{t+1}-e_t=-\Delta'(t)$. For F_s,

$$\Delta(z)=\omega\Big(\frac{1}{z+1}-\frac{1}{z+s}\Big)=\frac{\omega(s-1)}{(z+1)(z+s)},$$

and for F_χ, with $\omega=-\frac{1}{6}$,

$$\Delta(z)=\frac{1-z}{216\,(z+1)(z+\frac{1}{3})(z+\frac{1}{2})}.$$

Theorem 5.1 (selection)**.** *Let F be a frame of (2.1) with d digamma terms, and consider the alternating geometry of parameter c. There is K, depending only on d and on the period 2 of the border, such that for every large even r and every $x_0\in\mathbb{Q}$ some member $P_{r,I}$ with $|I|\le K$ satisfies $P_{r,I}(x_0)\neq 0$.*

Sketch of proof. Write $N=r+2m$ and $n=r+m$. By barycentric interpolation a vector in the right kernel of $\Phi_{r,\emptyset}(x_0)$ corresponds to a pair of polynomials (u,w) of degree less than n with the *value conditions* $w(t)=f_tu(t)$ at the N nodes and the *jet conditions* $w'(t)=f_tu'(t)+(x_0-e_t)u(t)$ at the common nodes t, and similarly for the left kernel. Two such pairs have zero cross product $w_1u_2-w_2u_1$: its degree is at most $2n-2$, while the value and jet conditions give $N+r=2n$ zeros counted with multiplicity.

For a pair whose degrees are small enough, the discrete Wronskian $W(z)=Q(z)[w(z+1)u(z)-w(z)u(z+1)]-A(z)u(z)u(z+1)$, where $\Delta=A/Q$ in lowest terms, has more prescribed zeros at the nodes than its degree allows, and so vanishes identically. Then w/u would be a rational antidifference of Δ, which is impossible: the simple pole of Δ at -1 is isolated in its class modulo $\mathbb{Z}$, because the other poles of Δ (at $-s$ for F_s, and at $-\frac{1}{3}$ and $-\frac{1}{2}$ for F_χ) are not congruent to -1 modulo $\mathbb{Z}$, and clearing denominators in $w(z+1)/u(z+1)-w(z)/u(z)=\Delta(z)$ would propagate a root of the denominator indefinitely in one direction. Counting the zeros by chains of nodes of step 2 bounds the dimension of the kernel, and so the corank of $\Phi_{r,\emptyset}(x_0)$, by a number K depending only on d and on the number of such chains, which is bounded independently of r.

Finally, the left and right kernels map to the same interpolation space, and at a common node t their coordinates are the values $u(t)$ times nonzero factors. No nonzero kernel pair has u vanishing at every common node (dividing by the node polynomial of the common nodes would give a pair of degree less than m satisfying the value conditions at all N nodes, and the

Wronskian argument applies again). Hence there is a set I of $|I| = \operatorname{corank}$ common nodes such that no nonzero left or right kernel vector vanishes on I. The rows outside I then span the row space, the columns outside I span the column space, and the principal minor deleting I is nonzero. □

The theorem applies to every frame of (2.1); for F_χ the pole of Δ at -1 is again isolated in its class modulo $\mathbb{Z}$, the other poles being $-\frac{1}{3}$ and $-\frac{1}{2}$. In the formal proof the depth is $K \le 64$ for all the frames of this paper.

6. Analytic decay

In this section we bound $|P_{r,I}(x)|$ for x near ξ. The argument is that of OpenAI [31] for square determinants, with three changes: the determinant has two families of nodes (rows and columns), the comparison measure is a density on finitely many segments rather than an explicit equilibrium measure, and the inequalities that remain are verified on a finite cover by cells. The result is one theorem for every geometry parameter $c \le 1$ and every frame of (2.1).

6.1. **A contour representation.** Hermite's formula gives $\mathsf{L}_F[t,t'] = \frac{1}{2\pi i}\oint F(z)v_t(z)v_{t'}(z)\,dz$ with $v_t(z) = 1/(z-t)$, the contour enclosing the nodes and no pole of F. Write $F = \sum_{u\in\mathcal{T}} w_u\psi(z+u)$, where $\mathcal{T}$ is the set of shifts of F and $\sum_u w_u = 0$: for F_s, $\mathcal{T} = \{1, s\}$ with the weights $w_1 = 1$ and $w_s = -1$, and for F_χ, $\mathcal{T} = \{\frac{1}{3}, \frac{1}{2}, 1\}$. The reflection formula $\psi(w) = \psi(1-w) - \pi\cot\pi w$ splits $F = \tilde{F} + E$, where $\tilde{F}$ is analytic in the left half-plane and $O(1/|z|)$ there, and

$$E(z) = -\pi \sum_{u\in\mathcal{T}} w_u \cot\pi(z+u)$$

is 1-periodic with $|E(z)| \le C_E e^{-2\pi|\operatorname{Im} z|}$ for $|\operatorname{Im} z| \ge 1$, because the weights sum to zero. For F_s, $E_s(z) = \pi[\cot\pi(z+s) - \cot\pi z]$. Let Γ be a polygonal path, symmetric under complex conjugation, which crosses the real axis at a point $x^* < 0$ that is not a pole of F (the poles are the points $-(j+u)$, $j \ge 0$, $u \in \mathcal{T}$), runs through the left half-plane and ends in two rays. Moving the contour to Γ picks up the residues of F at the poles in $(x^*, 0)$, and the part of the integrand with $\tilde{F}$ integrates to 0 over Γ. Hence, for all integers $t, t' \ge 0$,

$$\mathsf{L}_F[t,t'] = \int v_t v_{t'}\,dm, \tag{6.1}$$

where m is the complex measure with the mass $-\operatorname{res}_y(F)$ at each pole $y \in (x^*, 0)$ of F and the density $(2\pi i)^{-1}E(z)\,dz$ on Γ. This is a Gram representation with respect to a measure with finitely many atoms and an exponentially small density on Γ. After scaling by r, the atoms lie on a segment $[-r_0, 0]$ with density d, the number of digamma terms of F (two for F_s, three for F_χ). The separation condition on x^* is a finite check on the fractional parts of $u - x^*$, $u \in \mathcal{T}$.

6.2. **Two-family Andréief bound.** For a set $C'' \subseteq C$ of common nodes put $\mathcal{M}_{C''} = \omega\, \mathsf{L}_F[C'' \sqcup X', C'' \sqcup Y']$. For a member with common nodes $C' = C \setminus I$ put $R' = C' \sqcup X'$, $Q' = C' \sqcup Y'$ and $n = |C'| + m$. By multilinearity in the diagonal,

$$P_{r,I}(\xi + h) = \sum_{C'' \subseteq C'} h^{|C'|-|C''|} \det \mathcal{M}_{C''}. \tag{6.2}$$

Andréief's identity [14] for the two families $(v_t)_{t\in R'}$ and $(v_{t'})_{t'\in Q'}$ gives

$$\det \mathsf{L}_F[R', Q'] = \frac{1}{n!} \int \det[v_t(z_i)]\, \det[v_{t'}(z_i)] \prod_{i=1}^{n} dm(z_i),$$

and Cauchy's determinant evaluates both factors. Taking absolute values (see [37] for the logarithmic potential theory behind what follows),

$$|\det \mathsf{L}_F[R', Q']| \le \frac{1}{n!} \int \prod_{i<i'} |z_i - z_{i'}|^2 \prod_{\mathcal{F}\in\{R',Q'\}} \frac{\prod_{t<t'\in\mathcal{F}} |t'-t|}{\prod_{i,\, t\in\mathcal{F}} |z_i - t|} \prod_i d|m|(z_i).$$

The integrand is the exponential of minus a discrete logarithmic energy: the free points z_i repel each other with weight 1, each node family repels itself with weight $\frac{1}{2}$ and attracts the free points with weight 1, and the density $|E| \approx e^{-2\pi|\operatorname{Im} z|}$ acts as an external field. Because the rows and columns differ only in the interleaved border, the two node families have the same macroscopic density: after scaling by r, both are

$$\nu_c = \mathrm{Leb}|_{[0,1]} + \tfrac{1}{2}\mathrm{Leb}|_{[1,1+c]},$$

of mass $b = 1 + c/2$.

6.3. **The energy certificate.** For a regularisation radius $\rho > 0$ let $k_\rho(z, w) = -\frac{1}{2}\log(|z-w|^2 + \rho^2) \le -\log|z-w|$.

Lemma 6.1 (conditional positive definiteness)**.** *For points $z_1, \dots, z_n \in \mathbb{C}$ and real numbers $a_1, \dots, a_n$ with $\sum_i a_i = 0$, $\sum_{i,j} a_i a_j k_\rho(z_i, z_j) \ge 0$.*

Proof. Frullani's integral $\log y = \int_0^\infty (e^{-u} - e^{-yu})\, du/u$ $(y > 0)$ and $\sum_i a_i = 0$ give $\sum_{i,j} a_i a_j (-\log(|z_i - z_j|^2 + \rho^2)) = \int_0^\infty \frac{e^{-u\rho^2}}{u} \sum_{i,j} a_i a_j e^{-u|z_i - z_j|^2} du$, and Gaussian kernels are positive definite: expanding $e^{2u\langle z, z'\rangle}$ in powers and the binomial theorem write the inner sum as a convergent sum of squares. □

A *certificate measure* is a probability measure σ, symmetric under conjugation, carried by finitely many segments $[A_i, B_i]$ (with rational endpoints) contained in $\Gamma \cup [-r_0, 0] \cup [0, 1]$, on each of which it has the density $f_i(y)\, dy$ in the parameter $y \in [0, 1]$ of the segment, with f_i a polynomial with rational coefficients and $0 \le f_i \le M_i$ for a rational M_i. We write $\Theta_\sigma = \sum_i M_i/|B_i - A_i|$ for the sum of the bounds of its densities with respect to arclength. Comparing the free points and the deleted nodes with $b\sigma$ through Lemma 6.1

bounds the free–free energy from below by a sum of one-point terms and the constant $b^2 I(\sigma)$; Riemann-sum comparisons replace the node sums by the potential U^{ν_c} and the node self-energy by $I(\nu_c)$, up to $O(r \log r)$. With the external field one arrives at the function

$$g(z) = b\, U^{\sigma}(z) - U^{\nu_c}(z) + (1-\varepsilon)\pi |\operatorname{Im} z|$$

for a small fixed $\varepsilon > 0$, where $U^{\sigma}(z) = \int \log \frac{1}{|z-w|}\, d\sigma(w)$ is the logarithmic potential of σ, and U^{ν_c} that of ν_c. A certificate consists of σ and rational numbers m', $g_{\min}$, J, I_σ, I_ν such that

(a) $g \geq m'$ on Γ;
(b) $g \geq g_{\min}$ on $[0,1]$, the scaled position of the deleted common nodes;
(c) $J \leq \int_{-r_0}^{0} \min(g(x) - m', 0)\, dx$;
(d) $I(\sigma) \leq I_\sigma$ and $I(\nu_c) \geq I_\nu$.

Theorem 6.2 (decay and window)**.** *Let $0 < c \leq 1$, let F be a frame of (2.1) with d digamma terms whose poles are separated from x^*, and let $(\sigma, m', g_{\min}, J, I_\sigma, I_\nu)$ be a certificate for Γ and ν_c. Let $\rho > 0$, $\delta_\rho \geq \pi \rho\, \Theta_\sigma$, $\delta_{\mathrm{m}} > 0$ and $\delta_J > 0$ be rational, and put*

$$\delta = 3b^2 \delta_\rho + 4b^2 \delta_{\mathrm{m}} + 2d\, \delta_J .$$

If D and γ are rational numbers with

$$D \leq 2(b\, m' + d\, J) - b^2 I_\sigma + I_\nu - \delta, \qquad \gamma \geq 2(m' - g_{\min}),$$

then for every depth K, every member of the alternating geometry of parameter c and every real h with $|h| \leq e^{-\gamma r}$,

$$|P_{r,I}(\xi + h)| \leq \exp\bigl(-(D - o(1))r^2\bigr).$$

The loss δ comes from replacing σ by finitely many atoms in the comparison with the free points: the regularisation radius ρ of Lemma 6.1 and the mesh of the atoms lower m' by $b(\delta_\rho + \delta_{\mathrm{m}})$ and raise I_σ by $\delta_\rho + 2\delta_{\mathrm{m}}$, and Darboux sums for the integral in (c) lower J by δ_J; inserted in the expression for D, these changes give δ. The tolerances δ_{m} and δ_J may be any positive rational numbers, since the atoms can be taken as fine as needed. In the certificates of this paper $\rho = 10^{-12}$ and $\delta < 10^{-9}$.

The node energy $I(\nu_c)$ is a finite sum $\sum_k a_k H(y_k)$, $H(y) = \frac{1}{2} y^2 \log|y| - \frac{3}{4} y^2$, over the rational differences y_k of the endpoints of the two pieces of ν_c, and the lower bound I_ν is checked with rational enclosures of the logarithms (by the series of artanh and a reduction to $[1,2)$). The energy $I(\sigma)$ is a finite sum of pair energies of polynomial densities on segments, each a closed form of the type polynomial times logarithm plus arctangent, bounded in the same way. The conditions (a)–(c) are statements about g on sets of positive measure, and are proved on a cover by cells.

6.4. **The cover by cells.** Each of the conditions (a)–(c) is checked on a finite cover by cells with exact rational arithmetic. On a cell, the potential U^σ is split into the contribution of the segments of σ near the cell and the rest. Near segments are bounded by a Taylor polynomial in the cell variable with an explicit remainder, or, for a cell that meets the line of a segment beyond its end, by the same bound for the segment with its orientation reversed; at a corner of Γ the two incident segments are treated together. Far segments are bounded by a multipole expansion with an explicit tail. The potential U^{ν_c} is an explicit elementary function and is bounded above by monotonicity on the cell. For (a) the cells are intervals of the arclength on each edge of Γ, and the bound is proved on a horizontal neighbourhood of the edge of fixed width; beyond the last vertex the ray is handled by an analytic tail bound. For (b) the cells are intervals of $[0,1]$. For (c) the integrand is bounded above on each cell of $[-r_0,0]$ by a constant or by a chord, and J is the sum of the integrals of these majorants.

Proposition 6.3 (certified constants)**.** *For each row of Table 2 there is a certificate measure σ on a polygon Γ, with a cover by cells accepted by the checks above, for which Theorem 6.2 gives the constants D and γ of the table.*

All inequalities are verified by the Lean kernel (§11); the certificate data (the polygon, the density σ and the cells) were computed with floating-point optimisation and interval arithmetic in Arb [18], and are checked in Lean with rational arithmetic only. The data play no role in the proof beyond being accepted.

Remark 6.4 (how good a certificate can be)*.* For a multiplier B the certificate meets the exponent B exactly when $B(D-\Lambda)-\gamma \ge 0$. This quantity is a concave function of σ, and its maximum over all probability measures on $\Gamma \cup [-r_0,0] \cup [0,1]$ is attained at the measure σ^* for which $b\sigma^*$ is the weighted equilibrium measure of the constrained energy problem on this set with the external field $(1-\varepsilon)\pi|\operatorname{Im} z| - U^{\nu_c}$ and the upper constraint $d\,dx$ on the segment [37]. Weak duality expresses the deficit of any certificate as a sum of nonnegative terms, the first of which is the energy of $2Bb(\sigma-\sigma^*)$, the difference between $2Bb\sigma$ and the optimal dual measure $2Bb\sigma^*$. Thus a certificate close to the best possible one must be close to σ^* in energy, and $g-m'$ is then a small ripple on the support of σ^*; this is why a pointwise verification on a cover is needed. This remark is not used in the proof of Theorem 1.1 and is not part of the formalization.

6.5. **Node removal and the window.** The argument applies to every subset $C'' \subseteq C'$ of the common nodes: removing common nodes turns them into points of the configuration that are not integrated over, and only common nodes can be removed, because a node removed from one family only would leave an uncancelled term of weight $\frac{1}{2}$. By condition (b) one obtains

$$|\det \mathcal{M}_{C''}| \le \exp\bigl(-(D-o(1))r^2 + \gamma r\,|C' \setminus C''|\bigr),$$

uniformly in C''. Inserting this into (6.2) with $|h|e^{\gamma r} \le 1$ gives Theorem 6.2: the sum has at most $2^{|C'|} = e^{O(r)}$ terms, each at most $e^{-(D-o(1))r^2}$. The deletion of at most K common nodes in a member changes the energy by $O(r)$.

7. The number $L(2, \chi_{-3})$

For the frame F_χ of (2.1) we use the alternating geometry of parameter $c = \frac{1}{2}$, so that $\alpha = \frac{1}{4}$ and $m = \lceil r/4 - \frac{1}{2} \rceil$, at all even scales r. The decay and the nonvanishing are given by the general Theorems 6.2 and 5.1; what changes is the content. The frame has three digamma terms, so at a prime p the increments $g(k)$ of §3 have poles in three residue classes, and the local bounds of Section 4 are expressions in two first poles instead of one. We first describe these poles, then the bounds that we use for F_χ, and then the constants.

7.1. **The first poles.** Let $p > 3$ be a prime. The increment $g(k) = \frac{1}{3k-2} - \frac{1}{2k-1} + \frac{1}{6k}$ has a pole modulo p exactly when p divides k, $3k-2$ or $2k-1$. Let k_A and k_B be the least $k \ge 1$ with $p \mid 3k-2$ and with $p \mid 2k-1$. Then

$$k_B = \frac{p+1}{2}, \qquad k_A = \begin{cases} \frac{p+2}{3}, & p \equiv 1 \pmod 6, \\ \frac{2p+2}{3}, & p \equiv 5 \pmod 6, \end{cases}$$

and the residues of $p\,g(k)$ at k_A and k_B are 1 and -1 if $p \equiv 1 \pmod 6$, and $\frac{1}{2}$ and -1 if $p \equiv 5 \pmod 6$. In the first case $k_A < k_B$ and the two residues cancel: the partial sums $\sum_{k \le t} g(k)$ are p-integral for $t < k_A$, and again for $k_B \le t < p$.

Both first poles are affine in one index. Define k_p by $6k_p = jp + 5$ with $0 < j < 6$; then $j = 1$ for $p \equiv 1 \pmod 6$ and $j = 5$ for $p \equiv 5 \pmod 6$, $k_p = \frac{j}{6}p + O(1)$, and

$$(k_A, k_B) = \begin{cases} (2k_p - 1,\ 3k_p - 2), & p \equiv 1 \pmod 6, \\ (2k_p - p - 1,\ 3k_p - 2p - 2), & p \equiv 5 \pmod 6. \end{cases}$$

Thus j plays the role of the root class of §4, with q replaced by 6.

7.2. **The local bounds.** Fix a prime $p > 6$ with $6N < p^2$, where $N = r + 2m$; every pole below N is then simple. Call a node t *regular* if $t < \min(k_A, k_B)$, or if $p \equiv 1 \pmod 6$ and t is a border node with $k_B \le t < p$. At a regular node the partial sum of the increments up to t is p-integral, and when t is a common node so is its diagonal entry.

Proposition 7.1. *Every member of the alternating geometry of parameter $\frac{1}{2}$ for F_χ satisfies $e_p(r, I) \le \tau$ for each of the following τ.*

(i) The bounds (T1) and (T1′): $2(r+m)$ and $2r + 2m - \|m\|_p$.

(ii) If $N \le 2p$: $2r + 2m - n_{\mathrm{reg}}$, where n_{reg} is the number of pairs formed by a regular node t and a side (row or column) on which t lies, such that $t + p$ is not a node of the other side.

(iii) *If* $N < \min(k_A, k_B)$*:* 0.
(iv) *The bound* $2r + 2m - (t_2 - t_1)$ *of Theorem 4.7, for every* n *and every window* $[t_1, t_2)$ *as there, with the cuts* $t \equiv 0$, $t \equiv k_A$, $t \equiv k_B$ *and* $t \equiv r \pmod p$.

Proof. (i) is proved as in §4.3.1. For (ii) we apply Lemma 4.1 with the following potentials. At a node that is not regular, the potential is 1 on its row and on its column. At a regular node t, the potential on its row is 1 if $t + p$ is a column node and 0 otherwise, and the potential on its column is 1 if $t + p$ is a row node and 0 otherwise. Their sum is $2(r + m) - n_{\mathrm{reg}}$, and they dominate the weights W. Indeed, a diagonal weight is 0 at a regular common node and at most 2 in general. Two nodes that are not congruent modulo p have weight at most 1, since all poles are simple, and weight 0 if both are regular. Two distinct congruent nodes have weight at most 2. Since $N \le 2p$ and regular nodes are smaller than p, at most one of them is regular; if the row node t of such an entry is regular, its column node is $t+p$, so the potential on the row of t is 1, and symmetrically for a regular column node. For (iii), if $N < \min(k_A, k_B)$ then $N < p$, no index below N is a pole and no two nodes are congruent modulo p, so every entry is p-integral. For (iv), the proof of Theorem 4.7 uses the frame only through its poles: the step function V and the diagonal residues are constant between consecutive cuts, which is what the translation between the fibres of the window needs. □

The bounds (T2)–(T5), (LP1), (LP2), (R) and (S) of Theorem 4.5 are not used for F_χ.

7.3. **The content.** The checker of §4.5 and Theorem 4.14 apply to F_χ with q replaced by 6, the root class $j \in \{1, 5\}$ of the previous subsection, and the bounds of Proposition 7.1. These bounds are expressions of the class $\mathcal{E}$ in r, p, m, k_A, k_B; on each class of primes modulo 6 the substitution above makes them expressions in r, p, m, k_p, and their homogenisations are taken with $\kappa = j/6$. The proof of Theorem 4.14 uses the frame only through these bounds and the root class, and goes through with the bound (iii) in place of (T5) beyond the end of the table. The table of the formal proof starts at $\theta_0 = \frac{1}{68} = \alpha/17$, with the trivial bound $\ell_0 r$, $\ell_0 = \frac{5}{2} = 2 + 2\alpha$, below it. On $(\alpha/17, \alpha]$ it uses the bounds (i) and the windows (iv) with $1 \le n \le 16$; beyond α it uses (i), (ii), the windows (iv) with $n = 0$ between consecutive cuts, and the bound (iii), which ends the table. The checker accepts the table for the two classes $j = 1, 5$, and the content estimate holds for every even r and every member with $|I| \le 64$, with the content constant

$$\Lambda_\chi = 4.09592776731\ldots,$$

an explicit rational number (its numerator and denominator have 117 and 116 digits).

7.4. **Decay and nonvanishing.** The decay estimate is Theorem 6.2 for the frame F_χ, with $d = 3$ digamma terms and $c = \frac{1}{2}$. The node measure $\nu_{1/2}$ has mass $\frac{5}{4}$ and energy

$$I(\nu_{1/2}) = \frac{75}{32} - \frac{9}{8}\log\frac{3}{2} - \frac{1}{16}\log 2.$$

A certificate measure on a polygon, with a cover by cells accepted by the checks of §6.4, gives

$$D = \frac{4643511791526209475773}{10^{21}} = 4.64351\ldots,$$
$$\gamma = \frac{37521870783566737}{5\cdot 10^{15}} = 7.50437\ldots$$

Nonvanishing is Theorem 5.1 for F_χ, whose increment Δ has its pole at -1 isolated in its class modulo $\mathbb{Z}$; the depth is $K \le 64$. The deduction of Section 8, at the even scales, gives

$$\mu(L(2,\chi_{-3})) \le \frac{\gamma}{D-\Lambda_\chi} = 13.704516\ldots < 13.7046,$$

since $L(2,\chi_{-3}) = 4\xi_\chi$.

8. Proof of Theorem 1.1

The three estimates combine as follows.

Theorem 8.1 (deduction)**.** *Let F be a frame of (2.1) with its geometry and a depth K such that some member $P_{r,I}$, $|I| \le K$, is nonzero at every rational point for every large r (Theorem 5.1). Suppose that the content constant Λ of Theorem 4.14 and the constants D, γ of Theorem 6.2 satisfy $\Lambda < D$. Then $\mu(\xi) \le \gamma/(D-\Lambda)$.*

Proof. Let $\eta = D-\Lambda$, let $\mu_0 > \mu_1 > \gamma/\eta$, and suppose $|\xi - a/b| < b^{-\mu_0}$ with b large; put $L = \log b$. Let r be the least admissible scale with $r \ge \mu_1 L/\gamma$. By Theorem 5.1 with $x_0 = a/b$ there is a member with $P_{r,I}(a/b) \ne 0$. By Lemma 3.4, $|\operatorname{den}(P_{r,I})\,P_{r,I}(a/b)| \ge b^{-r} = e^{-rL}$. Write $a/b = \xi + h$ with $|h| < e^{-\mu_0 L} \le e^{-\gamma r}$ (as $\gamma r \le \mu_1 L + O(1) < \mu_0 L$ for large b). By Theorems 4.14 and 6.2, $|\operatorname{den}(P_{r,I})\,P_{r,I}(a/b)| \le e^{(\Lambda+o(1))r^2} e^{-(D-o(1))r^2} = e^{-(\eta-o(1))r^2}$. Comparing, $(\eta - o(1))r^2 \le rL$, whereas $\eta r \ge (\mu_1\eta/\gamma)L$ with $\mu_1\eta/\gamma > 1$; this is a contradiction for large b. □

Proof of Theorem 1.1. For each row of Table 1 the frame, the geometry parameter and the constants are those of Table 2. The content constant is given by Theorem 4.14 with the content table of the formal proof (for $L(2,\chi_{-3})$, by Section 7), the decay constants by Proposition 6.3, and nonvanishing by Theorem 5.1. In every row $\gamma/(D-\Lambda) < B$, and Theorem 8.1, applied at the even scales, gives $\mu(\xi) \le B$. Since ξ is a nonzero rational multiple of $\Psi(s,t)$ ((2.1), and $\xi_\chi = \frac{1}{36}\Psi(\frac{1}{3},\frac{2}{3})$), this proves Theorem 1.1. □

shifts	c	Λ	D	γ	$\gamma/(D-\Lambda)$	B
$\frac{1}{4}$, $\frac{3}{4}$	$\frac{1}{5}$	3.5149036	3.7447213	7.1904273	31.28751	31.2876
$\frac{1}{3}$, $\frac{2}{3}$	$\frac{3}{10}$	3.3104458	4.0291249	6.8272548	9.499725	9.4998
$\frac{k}{5}$	$\frac{2}{9}$	3.5128488	3.8135797	7.1655663	23.82717	23.8272
$\frac{k}{7}$	$\frac{1}{6}$	3.6391158643	3.6424792669	7.4259546075	2207.870	2207.8697
$\frac{1}{2}$	$\frac{3}{10}$	2.9923824	3.9785713	6.4350874	6.525207	6.5253
$L(2,\chi_{-3})$	$\frac{1}{2}$	4.0959278	4.6435117	7.5043742	13.70452	13.7046

TABLE 2. The data of the proof of Theorem 1.1: the shifts s of the frames F_s (and the frame F_χ), the geometry parameter c, the content constant Λ, the decay constant D, the window constant γ and the exponent. The constants are the rational numbers of the formal proof, rounded outward (Λ and γ up, D down) to the digits shown; the column $\gamma/(D-\Lambda)$ is the exact quotient, rounded up.

Remark 8.2 (relation to linear independence criteria). The deduction is a one-dimensional version of the familiar scheme behind the criteria of Nesterenko [29] and their use by Ball and Rivoal [2]: the nonzero integer $b^{|C\setminus I|}\operatorname{den}(P_{r,I})\,P_{r,I}(a/b)$ plays the role of a small linear form with integer coefficients. The difference is that the lower bound for the form comes from an arithmetic nonvanishing statement rather than from an analytic asymptotic, and that the size r of the determinant is chosen in terms of b.

9. The families

We collect the facts specific to each row of Table 2. In every case the frame is F_s of (2.1) with $s=k/q$, $\omega=q^{-2}$, and $\xi_s=-q^{-2}\Psi(\frac{k}{q},1)$; the content table, the certificate measure and the cover are the data of the formal proof.

9.1. **The shifts $\frac{1}{4}$ and $\frac{3}{4}$.** From $\psi'(\frac14)=\pi^2+8G$, $\psi'(\frac34)=\pi^2-8G$ and $\psi'(1)=\pi^2/6$,

$$\xi_{1/4}=-\frac{1}{16}\Psi(\tfrac14,1)=-\lambda_+,\qquad \xi_{3/4}=-\frac{1}{16}\Psi(\tfrac34,1)=\lambda_-.$$

The geometry parameter is $c=\frac15$, so $\alpha=\frac{1}{10}$. The two shifts $s=\frac{k}{4}$, $k=1,3$, have root classes $j\equiv kp^{-1}\pmod 4$, which exchange the two classes of primes modulo 4; by the transport of §4.5 they have the same content constant, and the same certificate serves both, since the decay depends on the shift only through the separation of the poles from the contour.

9.2. **The shifts $\frac{1}{3}$ and $\frac{2}{3}$.** Write $L=L(2,\chi_{-3})$. From $\psi'(\frac13)=9\sum_{n\ge0}(3n+1)^{-2}$, $\psi'(\frac23)=9\sum_{n\ge0}(3n+2)^{-2}$ and $\sum_{n\ge1}n^{-2}=\pi^2/6$ one gets

$$\psi'(\tfrac13)=\frac{2\pi^2}{3}+\frac92 L,\qquad \psi'(\tfrac23)=\frac{2\pi^2}{3}-\frac92 L \tag{9.1}$$

(the sum of the two is $\pi^2/\sin^2(\pi/3) = 4\pi^2/3$, and their difference is $9L$), hence

$$\xi_{1/3} = -\frac{1}{2}\Big(L + \frac{\pi^2}{9}\Big), \qquad \xi_{2/3} = \frac{1}{2}\Big(L - \frac{\pi^2}{9}\Big). \tag{9.2}$$

The geometry parameter is $c = \frac{3}{10}$. The content table uses the primes in the classes modulo 6; the two root classes $j = 1, 2$ correspond to first poles near $p/3$ and $2p/3$.

9.3. **The shifts $\frac{k}{5}$.** For $1 \le k \le 4$ the number $\Psi(\frac{k}{5}, 1) = \psi'(\frac{k}{5}) - \zeta(2)$ is a rational linear combination of the values $L(2, \chi)$ for the Dirichlet characters χ modulo 5, since $\psi'(\frac{k}{5}) = 25 \sum_{n \equiv k \pmod 5} n^{-2} = \frac{25}{4} \sum_{\chi} \bar{\chi}(k) L(2, \chi)$. The geometry parameter is $c = \frac{2}{9}$. The four members share one content table, indexed by the four root classes, and one certificate.

9.4. **The shifts $\frac{k}{7}$.** For $1 \le k \le 6$, likewise, $\psi'(\frac{k}{7}) - \zeta(2)$ is a combination of the values $L(2, \chi)$, χ mod 7, and of π^2. The geometry parameter is $c = \frac{1}{6}$ and there are six root classes. Here $D - \Lambda$ is small, about $3.4 \cdot 10^{-3}$, and the exponent is correspondingly large.

9.5. **The shift $\frac{1}{2}$.** Here $q = 2$, $\omega = \frac{1}{4}$, $\psi'(\frac{1}{2}) = \frac{\pi^2}{2}$ and $\psi'(1) = \frac{\pi^2}{6}$, so $\xi_{1/2} = -\frac{1}{4}\Psi(\frac{1}{2}, 1) = -\frac{\pi^2}{12}$. There is one root class, $j = 1$, with first pole $k_p = \frac{p+1}{2}$. The geometry parameter is $c = \frac{3}{10}$. As explained after Theorem 1.1, the resulting bound for $\mu(\pi^2)$ is weaker than the known ones.

10. Limits of the method

The exponents are determined by three constants through $\gamma/(D - \Lambda)$. Since $\partial\mu/\partial D = -\gamma/\eta^2$, the exponent is most sensitive to the decay where η is small: for $G \pm 5\pi^2/48$, $\gamma/\eta^2 \approx 136$, and for the shifts $\frac{k}{7}$ it is about $6.6 \cdot 10^5$. On the arithmetic side, the profiles $\mathcal{P}_j$ are minima of the bounds of Theorem 4.5, and every further local bound lowers Λ by the integral of the amount by which it lies below $\mathcal{P}_j$; the content constant of the method is the integral of the true exponent e_p, which the local bounds approximate from above. At third order this approximation is close to exhaustion for the shifts $\frac{k}{5}$ and $\frac{k}{7}$: in exact computations of the Smith normal forms at scales $r \le 2400$ and primes $p \le 2m$, the minimum of the local bounds, (U_n) included, equals the true exponent at all but a few isolated primes, and for the shifts $\frac{k}{7}$ the invariants of valuation at least 4 contribute about $5 \cdot 10^{-6}$ to Λ, against about $7 \cdot 10^{-4}$ gained by (U_n). The method's content at third order is thus essentially used up. These computations are not part of the proof. On the analytic side, Remark 6.4 explains why a certificate of the present form cannot do better than the constrained equilibrium measure on the chosen contour; a larger decay constant needs a different contour or a different treatment of the energy problem. The geometry parameter c trades the two against each other: a longer border increases D, through the

larger node measure, and Λ, through the border rows and columns of the assignment bound.

11. Formalization

The proof of Theorem 1.1, with every theorem, proposition and lemma it uses, is formalized in Lean 4 [10] with Mathlib [39]; the remarks and Section 10 are not part of the formalization. The development is available at

https://github.com/slitsey/catalan-irrationality-measure

and combines analytic and arithmetic proofs with generated rational data and kernel checks for the content tables of Section 4 and the decay certificates of Section 6; numerical computations alone are not used as proof. The constants are written out as series: G is $\sum_j (-1)^j/(2j+1)^2$, $L(2,\chi_{-3})$ is $\sum_n ((3n+1)^{-2} - (3n+2)^{-2})$, and $\psi'(\frac{k}{q}) - \zeta(2)$ is $\sum_{n\geq 0}(n+\frac{k}{q})^{-2} - \pi^2/6$. The final theorems, each stating that for every $e > 0$, all sufficiently large $b \in \mathbb{N}$ and all $a \in \mathbb{Z}$ coprime to b, $b^{-(B+e)} < |\xi - a/b|$, are the following (namespace `CatalanCore.Uniform.Bank`):

- `FourthsContent.Level3Bound.lambda_measure`: $G - 5\pi^2/48$ and $G + 5\pi^2/48$, $B = 31.2876$;
- `Chi3.KContent.chi3_measure_L2_k`: $L(2,\chi_{-3})$, $B = 13.7046$;
- `ThirdsContent.Level3Bound.thirds_plus_measure`, `thirds_minus_measure`: $L(2,\chi_{-3}) \pm \pi^2/9$, $B = 9.4998$;
- `FifthsContent.Level3Bound.fifths1_measure`, ..., `fifths4_measure`: $\psi'(\frac{k}{5}) - \zeta(2)$ for $k = 1,\dots,4$, $B = 23.8272$;
- `SeventhsContent.Level3Bound.sevenths_measure`: $\psi'(\frac{k}{7}) - \zeta(2)$ for $k = 1,\dots,6$, $B = 2207.8697$;
- `HalvesContent.Bound.pi2_measure`: π^2, $B = 6.5253$.

The shared results are (in the namespace `CatalanCore.Uniform`):

- `Content.BankShared.Check.hasScaledContent_of_check`: Theorem 4.14, with the local bounds of Theorem 4.5 as the catalogue (`Check/Catalogue.lean`, `Check/LP2W.lean`, `Check/Prefix.lean`, `Check/UF.lean`, the closed rank bound and the branches of Theorem 4.8), and `Content.BankShared.KTerm.hasScaledContent_of_kcheck`, its form for frames with several first poles (Section 7);
- `Content.BankShared.TwoTerm.LP2HypW.gle`: Theorem 4.6;
- `Content.BankShared.TwoTerm.Pre.prefix_gle`: Theorem 4.7;
- `Content.UFSimple.WrapCheck.catValid_wrap` and `Content.UFSimple.WrapPrefixCheck.catValid_prefix_full`: Theorem 4.8 for $n = 0$ and for $n \geq 1$, with Lemma 4.11 as `Content.UFSimple.WrapPrefixClosed.HSplit.det_ne` and `HSplit.dir`;
- `BlockGeom.hasNV0`: Theorem 5.1;

- `Decay.AltDecay.altGeom_hasScaledDecay`: Theorem 6.2, with the node energy check (`Decay/Nodes/EnergyCheck.lean`), the contour check (`Decay/ContourCheck.lean`) and the cell checks (`Decay/PolyWire3`);
- `measure_of_family_nv0_even`: Theorem 8.1 at the even scales.

The solution has no `sorry` and no `native_decide`, and the command `#print axioms` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The prime number theorem in arithmetic progressions enters the content asymptotics. The formal source for it is the PNT+ project [40], through OpenAI's workspace and compatibility additions, whose Wiener–Ikehara theorem we use.

The proofs were checked with Lean `v4.35.0-rc2` and the matching Mathlib release. The vendored files from OpenAI's repository and from the PNT+ project were converted to Lean's module system and ported to the current toolchain; the repository's README and provenance documents describe the changes.

Disclosure of AI use

The mathematics, the Lean formalization, the certificate generators and the text of this paper were produced by AI systems at the request of the author, who posed the problems, directed the work and is responsible for this paper: Claude Opus 5.5, Claude Sonnet 5.5 and Claude Haiku 5.5 (Anthropic), running in Claude Code, and GPT-6.1 Sol (OpenAI), running in Codex. The underlying irrationality proof for G and its Lean formalization are OpenAI's [31]. We use Lean 4 [10], Mathlib [39], the PNT+ project [40], and Arb [18] (through `python-flint`) for interval arithmetic in generating certificate data, which Lean re-checks. The formal verification in Lean is intended as an independent check of every step of the argument above.

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